

Plasmonic Nanostructure Lattices for High-Performance Sensing

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Plasmonic nanostructures show great promise for sensing because their nanoscale confined light fields are sensitive to the change in the surroundings. Conventional plasmonic sensors based on surface plasmon polaritons (SPPs) and localized surface plasmon resonances (LSPRs) have inspired considerable progress in sensing but still suffer from an oblique incidence or moderate sensitivity. This review focuses on how the rational design of novel plasmonic nanostructures can enable high-performance sensing. Patterned nanostructures such as nanoparticle (NP) lattices to support surface lattice resonances (SLRs) and plasmonic nanogaps with nanogap modes are emerging to overcome the sensing limitations of SPP and LSPR. Moreover, hybrid nanostructures of plasmonic components with functional materials, such as metal-organic frameworks, 2D materials, oxides, and polymers, show opportunities to further improve sensitivity and selectivity. In addition, plasmonic nanolasing and resonance modes from new materials exhibit appealing features for sensing. It is expected that further studies on plasmonic nanostructures with low-loss materials, chirality characteristics, novel devices, and advanced fabrications will provide outlooks for high-performance sensing.

nonlinear physics,^[3,4] chemistry,^[5] spectroscopy,^[6–8] and sensing applications.^[6,9–12] Compared to conventional fluorescence analysis, plasmonic sensors are advantageous with features such as label-free processing and robust platforms without blink or bleach.^[9] With the trend of miniaturized sensors for clinical evaluation such as real-time diagnosis and personalized care as well as lab-on-a-chip devices, particular attention has been devoted to highly sensitive and selective sensing based on plasmonic sensors.

Surface plasmons are collective oscillations of free electrons at the interface between plasmonic and dielectric materials.^[13,14] Depending on supporting structures and propagating features, plasmonic resonance can be categorized into two basic modes including surface plasmon polaritons (SPPs) and localized surface plasmon resonances (LSPRs).^[1] SPPs are evanescent waves propagating along with the interface between the dielectric materials and metallic thin films, which can be

1. Introduction

Plasmonic nanostructures can squeeze light into sub-wavelength volumes by supporting surface plasmons.^[1,2] This appealing light confinement at the nanoscale has triggered intense research activities aimed at exploring plasmonic nanostructures for

excited through prism coupling, grating coupling, or other coupling mechanisms.^[14] SPPs have greater momentum than that of a free photon at the same frequency, thus, light directly incident from the air will not couple to the plasmon, and momentum-matching techniques are required. While LSPRs are nonpropagating resonances from the coupling of electron oscillation and electromagnetic field at the NP surface. Compared to a typical spectral width of ≈ 50 nm for SPPs, the resonance width of LSPRs normally exceeds ≈ 80 – 100 nm.^[15] Both SPPs and LSPRs are sensitive to the change in the surrounding refractive index (RI) that leads to spectrum modification (**Figure 1a**). Despite their promise, sensing based on SPPs requires angle alignment components to realize momentum match for propagating features, which limits the simplification and miniaturization of their devices. LSPR-based detections can serve as high-throughput screening and highly integrated sensing platforms, but broad resonance widths substantially hinder their promise for highly sensitive sensing. Basic plasmonic modes suffer from intrinsically high ohmic losses of metals resulting in low-quality factors and consequent low sensitivities. Numerous efforts have been dedicated to improving the sensitivity and selectivity of sensing based on novel plasmonic structures. We summarize these strategies into four categories, 1) designing ordered nanostructures and nanogaps to couple SPP or LSPR modes with other photonic modes (**Figure 1b**), 2) constructing hybrid structures consisting

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DOI: 10.1002/adom.202300401

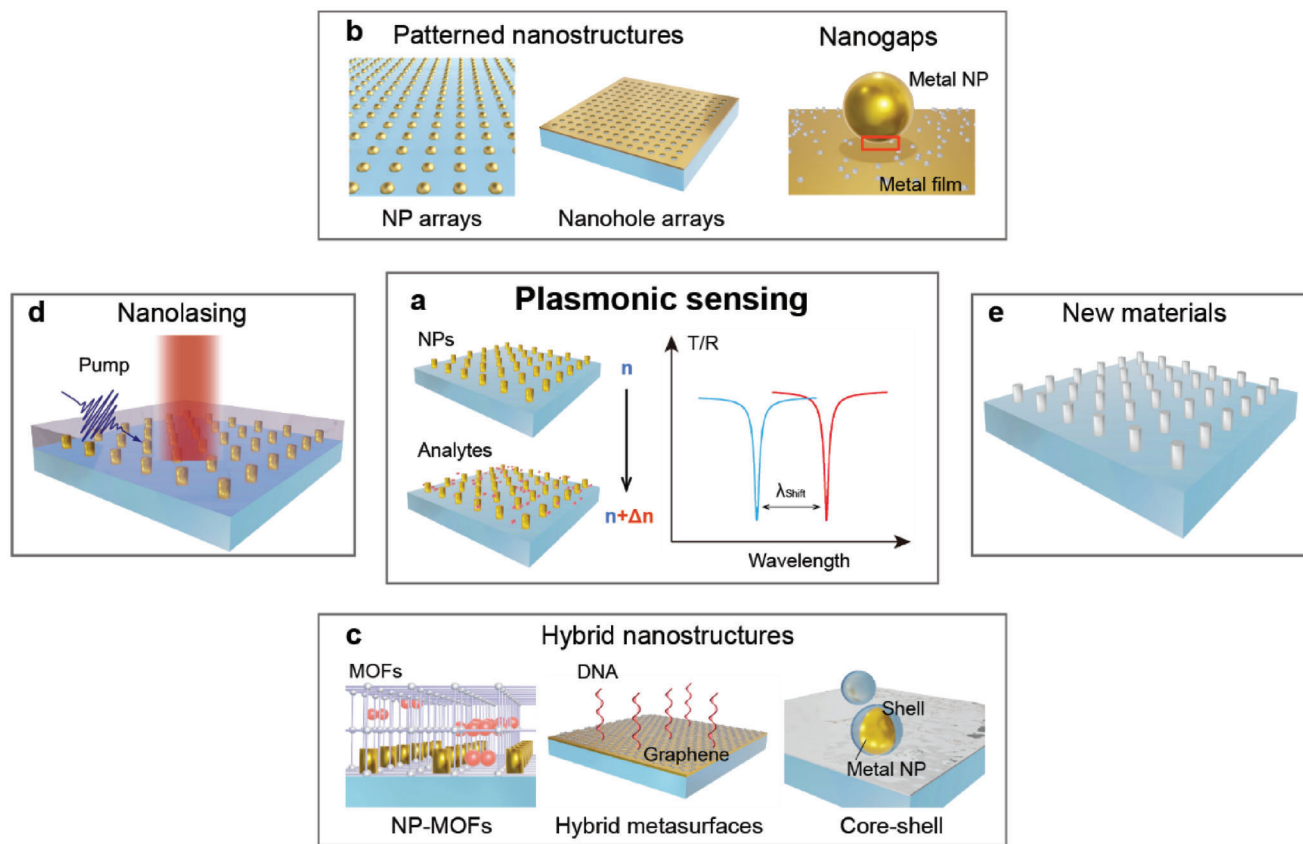


Figure 1. Plasmonic nanostructure lattices for high-performance sensing. a) Fundamentals and mechanism of plasmonic sensing. b) Patterned nanostructure lattices (e.g., NP arrays and nanohole arrays) and nanogaps for plasmonic sensing. c) Hybrid nanostructure lattices with functional materials, including NP-MOFs, hybrid metasurfaces, and core-shell structures for sensing. d) Plasmonic nanolasing for high-sensitivity sensing. e) New plasmonic materials beyond noble metals for sensing.

of plasmonic and functional materials (Figure 1c), 3) enhancing plasmonic feedbacks by nanolasing (Figure 1d), and 4) utilizing plasmonic modes in materials beyond noble metals (Figure 1e).

In this review, we focus on the novel design of plasmonic nanostructures that is beyond basic SPP and LSPR modes for sensing with high sensitivity or selectivity. We concentrate our discussion most on nanostructures with lattice arrangement and emerging nanostructures that can be readily incorporated into lattices because we believe the lattice arrangement is critical for practical and reproducible sensing applications. In Section 2, we introduce the fundamentals and properties of SPP and LSPR modes for sensing as well as parameters for sensing performance evaluation. In Section 3, we discuss the fabrication of the NP lattices, ordered plasmonic nanostructures that support coupled SPP or LSPR/photonic modes such as NP arrays, nanogroove arrays, nanohole arrays, and nanogaps in sensing applications. In Section 4, we review sensors based on the hybridization of plasmonic structures with functional materials including metal-organic frameworks (MOFs), 2D materials, dielectrics, and polymers. In Section 5, we concentrate on plasmonic nanolasing for high-sensitive sensing. In Section 6, we summarize the recent development of materials beyond traditional noble metals, such as alloys, heavily doped semiconductors, and 2D materials. Further, we conclude this review with emerging research areas in-

cluding new resonance modes, chirality, device innovation, and fabrication advances for highly sensitive sensing with plasmonic nanostructures.

2. Fundamentals of Plasmonic Sensing

Plasmonic nanostructures have been effective building blocks for label-free sensing due to their simple mechanism, low cost, fast detection, and real-time monitoring capability.^[9,16–18] Plasmonic modes respond to the change of RI with spectral modifications, for example, biomolecules with higher RI than solutions can lead to redshifts of the resonance wavelength when molecules approach the plasmonic surface.^[19–21] This section overviews the fundamentals of two basic modes and summarizes the definition of evaluation parameters for sensing performance.

2.1. Surface Plasmon Polariton

Surface plasmon polariton, also known as propagating surface plasmon, is an evanescent surface electromagnetic wave that propagates at the metal-dielectric interface (Figure 2a).^[1,14] SPP waves are bounded to the collective motion of mobile charges

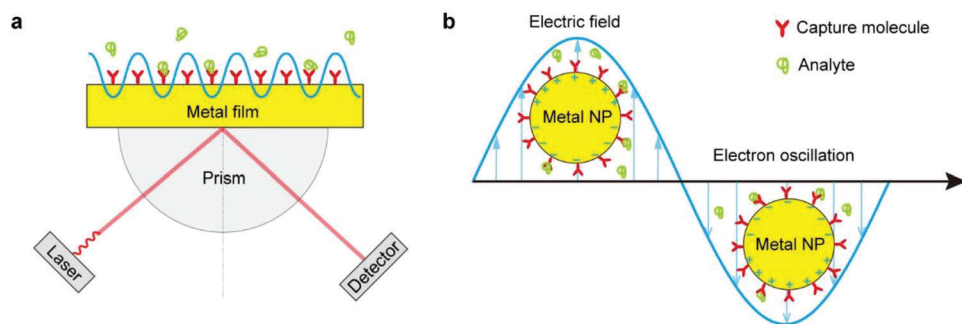


Figure 2. Fundamentals of plasmonic sensing based on SPP and LSPR modes. a) Plasmonic sensing based on SPP modes, where SPPs are generated from the interface between the metal and dielectric with laser pumped. When analytes move to the surface of the metal (with or without bonding to capture molecules), the wavelength dip or the phase changes (for a fixed wavelength). b) Plasmonic sensing based on LSPR modes from metal NPs.

with subwavelength light confinements and near-field enhancement. Due to the coherent interaction between SPP waves and electrons, the propagation constant of SPP waves is greater than the wavenumber of light in the dielectric. Thus momentum matching methods such as prism coupling,^[22,23] and grating coupling (e.g., nanohole arrays)^[13,24,25] are required to excite the SPP. The most common structure for SPP excitation is the Kretschmann configuration, where the metal film is coupled with a glass prism.^[22] According to Maxwell equations and boundary conditions at the interface of metal and dielectric, the dispersion relation of SPP can be described as

$$k_{spp} = k_0 \sqrt{\frac{\epsilon_d \epsilon_m}{\epsilon_d + \epsilon_m}} \quad (1)$$

Where k_{spp} is the wavevector of SPP, k_0 is the wavevector in free space, ϵ_d and ϵ_m are the permittivities of the dielectric and the metal, respectively.^[1,26,27] The momentum mismatch between the incident light and propagating waves is overcome with $\epsilon_d + \epsilon_m = 0$, and SPPs are excited at the interface. The excitation of SPP is accompanied by the appearance of a minimum in the reflectance spectrum at a specific wavelength and angle of incidence.^[28,29] SPP structures have led to a major advance in the development of label-free, real-time sensing on biomolecules, explosives, and environmental pollutants.^[21] However, SPP sensors show low selectivity for small analytes because their decay length is in the order of hundred nanometers. Further, the minimization of SPP devices is hindered due to the requirement of an angle alignment component for momentum-matching conditions.

2.2. Localized Surface Plasmon Resonance

Localized surface plasmon resonance is the nonpropagating resonance of the collective oscillation of free electrons in metal nanostructures (Figure 2b).^[6,30] Compared to the SPP, LSPR can be excited in a metal NP by the incident light without momentum matching owing to its lack of translational symmetry. Metal NPs can be precisely fabricated by colloidal chemistry and lithography techniques,^[31,32] thus LSPRs are suitable for miniaturized sensing to achieve multiplexed and high-throughput screening platforms. For a typical homogenous spherical metal NPs with a

size much smaller than the light wavelength, the scattering cross-section C_{sca} and absorption cross-section C_{abs} can be calculated to^[1]

$$C_{sca} = \frac{8\pi}{3} k^4 a^6 \left| \frac{\epsilon_m - \epsilon_d}{\epsilon_m + 2\epsilon_d} \right|^2 \quad (2)$$

$$C_{abs} = 4\pi k a^3 \text{Im} \left[\frac{\epsilon_m - \epsilon_d}{\epsilon_m + 2\epsilon_d} \right] \quad (3)$$

Where k is the wavenumber of the incident light, a is the radius of metal spherical NP. C_{sca} and C_{abs} reach the maximum at the minimum of $|\epsilon_m + 2\epsilon_d|$, then the extinction dips are generated in the spectrum accompanied by a strong near-field enhancement. The frequency and intensity of resonances depend on the material composition, size, shape of NPs, and the permittivity of the surrounding environment.^[33,34] LSPR modes can convert changes in the surrounding RI into instant modifications of the extinction spectrum, and real-time monitoring of dynamic variation around NPs can be achieved by integrating with high-resolution spectrometry. However, the moderate intensities and broad linewidths of LSPRs from the high radiative loss restrict their performance for highly sensitive sensing.^[35]

2.3. Parameters for Performance Evaluation

The wavelength shift of a resonance peak ($\Delta\lambda$) is linear with RI change (Δn), and the sensitivity (S) is expressed by the change of wavelength to the unit refractive index (nm/RIU),

$$S = \frac{\Delta\lambda}{\Delta n} \quad (4)$$

The figure of merit (FOM) is a frequently used parameter to evaluate the performance of sensing,^[27] and $\Delta\omega$ is the full width at half maximum (FWHM) of a plasmonic resonance,

$$\text{FOM} = \frac{\Delta\lambda}{\Delta n} \cdot \frac{1}{\Delta\omega} \quad (5)$$

For a complex plasmonic resonance peak without a single-line shape, wavelength change is replaced by the relative change in

intensity $\Delta I/I$ at a fixed wavelength λ that results from Δn , and a modified FOM (FOM*) can be expressed as^[36]

$$\text{FOM}^* = \frac{\Delta I}{I \Delta n} \quad (6)$$

The quality factor ($Q = \frac{\lambda}{\Delta\omega}$) is linked to the ratio of energy stored to the energy lost by an oscillator, which represents the ability of a plasmonic nanostructure to store energy.^[15] The limit of detection (LOD), defined as the measurable lowest concentration of analytes with a signal larger than 3σ (σ is the signal noise), is used to assess the highest detection capability of the sensor.^[15,37–38]

3. Patterned Nanostructures and Nanogaps

The sensing performance and detection limit are restrained by the FWHM of resonances, which is determined by the radiative loss and nonradiative loss.^[39] The nonradiative loss is intrinsically generated by the electron scattering within the materials, while radiative loss originates from the coupling of oscillations to the far field.^[40–42] Therefore, an effective way to improve the sensitivity and detection limit of plasmonic sensing is to reduce the FWHM of the resonance by organizing the plasmonic nanostructures into patterned lattices. In this section, we briefly summarize the typical methods for the preparation of the plasmonic NP lattices including both top-down lithography methods and bottom-up assembly methods. Further, we focus on how the patterned plasmonic nanostructure lattices and nanogaps with different resonance modes can achieve narrow linewidths for highly sensitive sensing.

3.1. Fabrication of Patterned NP Arrays

Due to the nanoscale feature size, traditional photolithography approaches are not suitable for the typical plasmonic nanostructure lattices. Various fabrication methods have been developed to generate pattern arrays, which can be summarized into two categories: top-down lithography methods and bottom-up assembly methods. Traditional top-down methods can produce highly ordered arrays with diverse shapes of units; however, high-throughput and cost-effective fabrication processes are desired for practical sensing applications where bottom-up assembled arrays may be more favorable.

Electron beam lithography (EBL) is the most used method to prepare plasmonic nanostructure lattices because of its high-resolution and mask-free nature (Figure 3a). EBL method can produce arbitrary shapes of units with resolution down to a few nanometers but is limited by its micrometer scale writing area, long writing time for large-scale lattices, stitch error, and high cost.^[43–48] Similar to EBL, focused ion beam (FIB) milling is a convenient tool for the direct formation of plasmonic arrays by milling out the target patterns from thin films (Figure 3b).^[49–53] Compared to direct pattern generation on targeted substrates in EBL and FIB methods, nanostencil and nanosphere lithography can generate periodic plasmonic arrays by employing hole arrays or polymer nanosphere monolayers as metal deposition masks.^[54–59] Nanosphere lithography has been a widely

used low-cost technique to fabricate large-area periodic nanostructures but is limited by the irregular morphology, connected units, and sharp tips because of the side deposition of metals (Figure 3c).^[60–62] Nanoimprint lithography is another high-throughput patterning technique for the fabrication of plasmonic NP lattices (Figure 3d).^[63–67] By applying pressure under high temperatures or UV lights, the periodic array structures can be transferred from the masks to the resists and further generate NP lattices on target substrates. Nanoimprint methods are advantageous for large and uniform arrays with simple unit structures but are restricted by their structural tunability because the patterns in the imprint masks are typically created by other pattern-generation methods.^[67]

Besides top-down approaches, the assembly of colloidal NPs into periodic arrays provides rapid and low-cost methods to form large-scale plasmonic NP lattices.^[68] The NP arrangement can be achieved by chemical interaction, electrostatic attraction, capillary force assembly, dielectrophoresis and electrophoretic deposition, polymer shell assembly, and template-induced colloidal self-assembly.^[69–78] Chemical functionalization on the surface can introduce specific interactions between NPs and substrates, which can be employed to form NP lattices. Particularly, nucleic acids and hybridization of DNA strands have shown great potential for NP assembly, 2D NP lattices and even vertical stacking NP lattices can be generated (Figure 3e).^[79,80] Similar to chemical interactions, NPs can also be arranged on substrates into lattices by taking advantage of opposite charges in the electrostatic processes (Figure 3f).^[69] The electrical driving force for NP assembly can also induce from the conductive surface through electrophoretic/dielectrophoretic processes.^[69,81,82] For example, vertically oriented, Au nanorods can be arranged into arrays due to the electric dipole moment induced along the rod by the electric field (Figure 3g).^[78] Further, combining the meniscus motion of the top plate and solvent evaporation of heated substrates, capillary assembly of NPs can be realized after NPs escape from the colloidal meniscus at the solution/air interface and get trapped in trenches of a low-wetting substrate (Figure 3h).^[81,83] Notably, these assembly methods are involved with pre-patterned substrates through top-down lithographic techniques. Direct formation of plasmonic NP lattices on nonpatterned substrates are advantageous for further simplifying the fabrication process and reducing the cost. Centimeter-scale hexagonal plasmonic NPs on glass substrates can be generated by colloidal self-assembly through the coating of a polymer shell around NPs and the interparticle distance can be tuned by changing the thickness and deformability of the polymer shells (Figure 3i).^[84] Template-induced colloidal self-assembly of NPs also shows potential for large-scale fabrication of the 2D arrays (Figure 3j).^[82,85–90] The optical qualities of assembled arrays are typically low because of the poor control of the number of NP per unit and arrangement defects (Figure 3k) but can be improved through thermal annealing posttreatments (Figure 3l).^[82,85]

3.2. NP Arrays Supported SLRs

Surface lattice resonances can be generated by coupling LSPR to the diffracted waves from arranged metal NPs in periodical arrays.^[15] Each NP oscillates in phase at resonance and forms

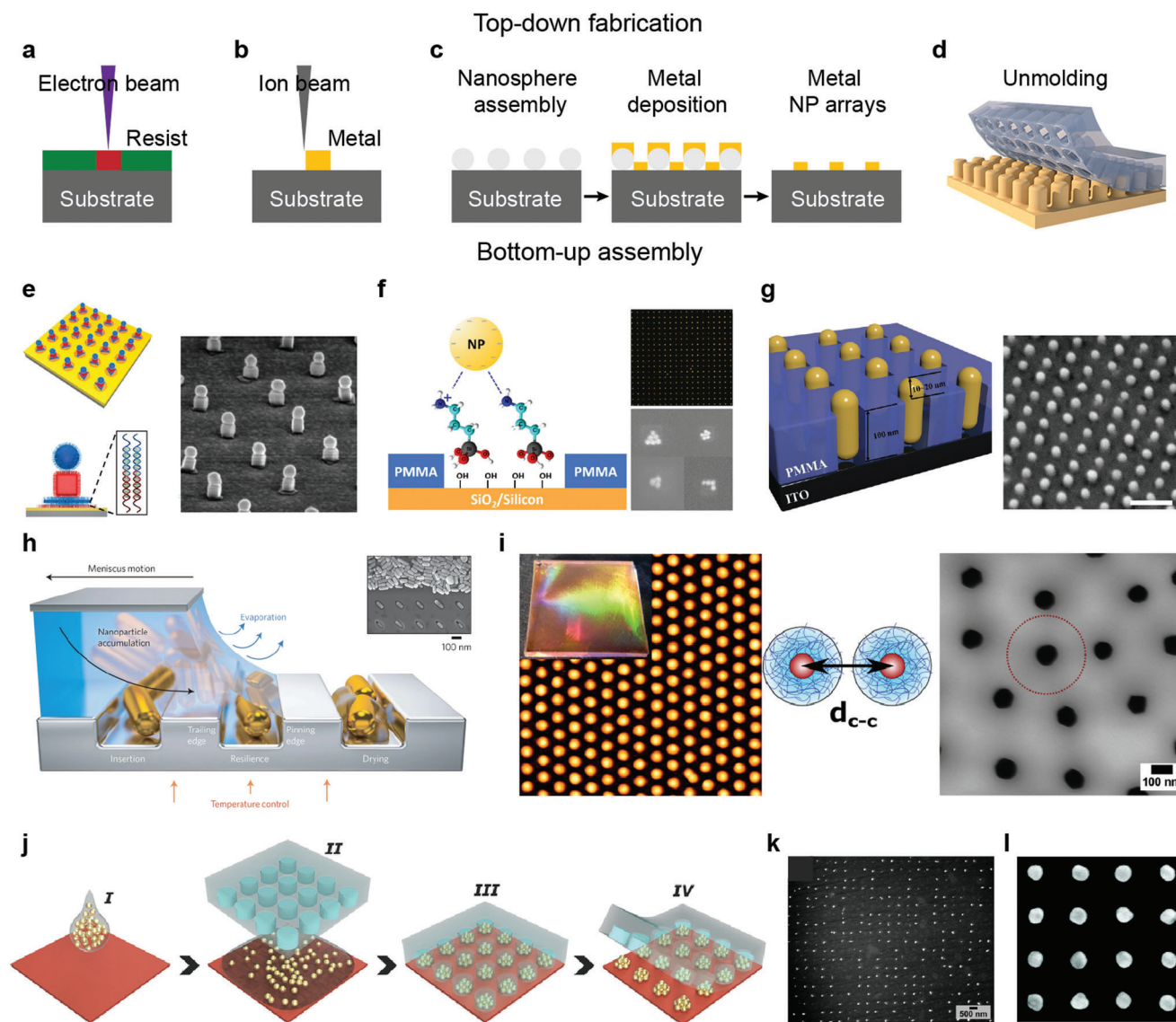


Figure 3. Top-down nanofabrication and colloidal assembly of NP arrays. a–d) Schematics of EBL, FIB, nanosphere lithography, and nanoimprint lithography processes. e) Schematic and SEM image of DNA-functionalized colloidal NPs of controlled size and shape assembled in a layer-by-layer manner. f) Schematic, dark field, and SEM image of directed chemical assembly of NPs on the substrate. g) Schematic and SEM image of the geometry of the vertically assembled gold nanorod array through the electrophoretic process. h) Schematic of the capillary assembly of NPs onto topographical traps of a low-wetting substrate by the capillary assembly. i) Photo, SEM, and TEM images of self-assembled Au NP arrays by coating polymer shells. j) Schematic of the fabrication process of NP arrays through template-induced colloidal self-assembly. SEM images of k) direct growth of NPs on the substrate and l) NP arrays of template-induced colloidal self-assembly with postannealing treatment. d) Reproduced with permission.^[67] Copyright 2021, ACS. e) Reproduced with permission.^[79] Copyright 2018, AAAS. f) Reproduced with permission.^[69] Copyright 2018, ASC. g) Reproduced with permission.^[78] Copyright 2020, Wiley-VCH. h) Reproduced with permission.^[83] Copyright 2016, Springer Nature. i) Reproduced with permission.^[84] Copyright 2020, ACS. j) Reproduced with permission.^[88] Copyright 2021, Wiley-VCH. k) Reproduced with permission.^[85] Copyright 2022, Wiley-VCH. l) Reproduced with permission.^[82] Copyright 2022, RSC.

standing waves across the lattice. Instead of radiating into far fields, the scattered light is recaptured for excitation of adjacent NPs to suppress the radiative loss with linewidths down to 1–2 nm (**Figure 4a**).^[15] The increased quality factor and narrow resonance linewidths lead to potentially higher sensitivity and lower LOD in sensing than sensors based on individual NPs.

SLRs can be tuned from UV to mid-infrared wavelength by rationally designing the unit composition, size, array arrangement,

refractive environment (symmetry/asymmetry of refractive index), and substrate conductivity.^[15] Depending on the geometry of charge oscillations, resonance modes of SLR can be classified as dipolar resonance, quadrupolar resonance, and high-order hybrid resonance (**Figure 4b**).^[91–93] For example, the quadrupolar resonance can be supported by large NPs in lattices.^[91,92] In Al NP hexagonal arrays with large NP diameters (e.g. diameter = 180 nm with periodicity = 346 nm), the quadrupolar mode

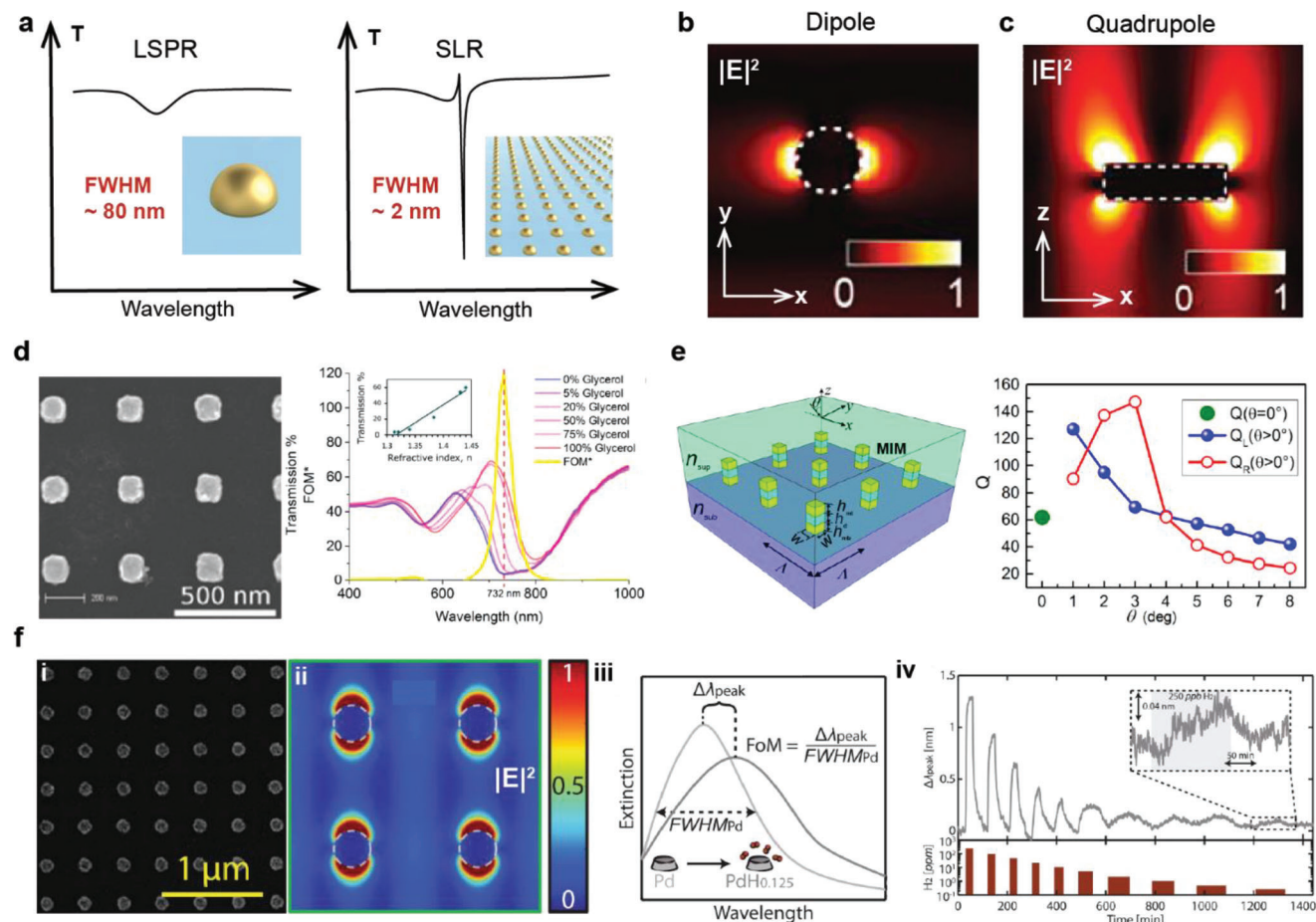


Figure 4. Plasmonic sensing based on SLR modes. a) Examples of transmission spectra of the LSPR and SLR modes. The FWHM of a typical LSPR is ≈ 80 nm and the FWHM of an SLR can be down to 1–2 nm theoretically. b) Simulated near-field distribution of the SLR with a dipolar mode. c) Simulated near-field distribution of the SLR with quadrupolar modes (Al NP array, diameter = 180 nm, and periodicity = 346 nm). d) SEM image of Au square arrays with the period of 400 nm (left) and transmission spectra at different glycerol concentrations (from 0% to 100%) with the highest FOM* of 120 (right). e) Schematic of a metal-insulator-metal (MIM) array (left) and quality factors at different incidence angles (right). f) Pd NP arrays for hydrogen sensing. SEM image i) and simulated near-field distribution ii) of Pd NP arrays (diameter = 180 nm, height = 45 nm, periodicity = 550 nm, and thickness_{PMMA} = 200 nm). iii) Schematic of working principle for hydrogen sensing. iv) $\Delta\lambda_{\text{peak}}$ response to decreasing hydrogen concentration (1000 ppm to 0.25 ppm) in Ar atmosphere with a LOD of 250 ppb. (b) Reproduced with permission.^[93] Copyright 2017, Elsevier Ltd. (c) Reproduced with permission.^[91] Copyright 2016, NAS. (d) Reproduced with permission.^[101] Copyright 2014, ACS. (e) Reproduced with permission.^[107] Copyright 2019, OAS. (f) Reproduced with permission.^[37] Copyright 2022, Springer Nature.

exhibits a different near-field pattern and narrow resonance linewidth compared to dipolar resonance (Figure 4c).^[91] The field enhancement is two to three orders of magnitude larger than the incident and hotspots reside at the opposite sides of NPs along the polarization direction.^[93] SLRs can also be categorized as in-plane and out-of-plane resonance depending on the NP diameter and height.^[91,94,95] Both in-plane resonance mode and out-of-plane mode can couple to the lattice structure leading to Fano resonances.^[95] Further, superlattice can generate a high-order diffractive mode of SLR with strong and tunable field enhancement (10- to 40-fold) induced by microscale periodic patch-to-patch coupling.^[96] NP arrays supporting SLR mode with narrow linewidth were demonstrated by both calculations and experiments for sensing.^[97–100] For example, Au NP arrays under both oblique incidence and normal incidence exhibit narrow resonance.^[98,100] In an asymmetric air/water environment, SLRs

from Au square NP arrays on a glass substrate with a FOM* of 120 provide access for biosensing with a wavelength range from 700 nm to 1000 nm (Figure 4d).^[101] However, quality factors of these lattices at normal incidence are only 19 in air, 45 in water, and 85 in glycerol respectively due to a nonhomogeneous RI environment.^[101] The quality of SLRs can be significantly affected by the refractive index environment of NP arrays, and the index match between substrate and superstrate is preferred for high-quality SLRs. However, it remains challenging to satisfy these index-match conditions for either bio- and gas sensing because the refractive indexes of aqueous solutions and gases are usually lower than that of the solid substrates (glass, quartz, etc.). The index-mismatch environment between the bottom substrate and the upper cladding (water/glass or air/glass interfaces) is unfavorable for the efficient coupling between NPs by the in-plane diffraction orders, which would broaden the resonance

linewidth, deteriorate the sensitivity and quality factor of sensing based on SLRs at normal incidence.^[15,101–103] Therefore, strategies to achieve high sensitivity under index mismatch conditions are favorable for sensing applications. Arrangement of metallic mirrors below the metal NPs and an attenuated total reflection (ATR) geometry are proposed to solve the issue of asymmetry index environment.^[104,105] However, both the mirror plane and the ATR geometry require a relatively large angle of the incidence, which leads to a complicated instrument setup.^[104,106] Narrow SLRs with high quality factors in an asymmetric environment were recently demonstrated under normal or small oblique incidence with small angles in arrays with metal-dielectric-metal (MIM) units (Figure 4e).^[107] These arrays support both in-plane dipolar and out-of-plane quadrupolar resonances.^[107] Under a refractive index constant of 1.0/1.52 air/polymer environment, quality factors reach 62 under normal incidence and 147 under oblique incidence (at $\theta = 3^\circ$) in the visible regime.

Importantly, the enhanced sensing from plasmonic lattices supporting narrow SLRs is not absolute. Improved sensing is due to the balance between the narrow resonance and strong field enhancement. This balance condition is naturally satisfied by sensing based on the detection of the RI change in the surrounding plasmonic units in lattices because of the redistribution of the electromagnetic field into the surroundings.^[15,108,109] However, the detection performance based on monitoring RI change inside the unit in lattices does not necessarily reach the best. Despite narrow linewidths of SLR, the removal of the electric field from metallic NPs is unfavorable for direct plasmonic sensors, such as Pd NP arrays for hydrogen sensing where dissociation of H_2 molecules at the surface of Pd NPs and the intercalation of H atoms into the metal lattice leads to changes in the dielectric function of Pd NPs and shifts plasmonic resonances of Pd NPs.^[37] Optimized Pd NP array structures by algorithmic technique to realize a balance between the narrow spectra feature and large field enhancement inside NPs has been proposed to realize hydrogen sensing with a LOD of 250 ppb (Figure 4f), which is about an order of magnitude higher than the previous detection limit.^[37]

3.3. NP Arrays Supported Resonance Modes Beyond SLRs

Besides SLR, other resonance modes supported by NP arrays such as SPP, LSPR coupled with Wood's anomaly (WA), and Fabry-Pérot (FP) mode exhibit narrow linewidth and high Q factor for sensing applications. There are growing efforts devoted to fabricating 2D periodic arrays for sensing, such as nanohole arrays,^[64,110] nanostrip arrays,^[111] and nanoslit arrays.^[112] 2D periodic hole arrays can excite SPP with small size for portable applications and broadband operation beyond the Kretschmann configuration.^[10,95] In these SPPs, the FWHM can be reduced by increasing the incidence wavelength or minimizing the size of the Rayleigh-like scatterer in the periodic nanostructure.^[113,114] A 2D Au hexagonal hole array with small Rayleigh-like scatterer sizes enables the FWHM down to 3 nm at the resonance wavelength of 960 nm and a large FOM of 730 (Figure 5a,b).^[115] Rayleigh-like scatters are nanoholes and nanoslits (highlighted with yellow circles and rectangles in Figure 4a) with feature sizes tuned by the exposure time. This SPP sensor exhibits a narrow

FWHM of less than 8 nm within a wide working wavelength from visible to near-infrared and the FOM of bulk sensing can reach as high as ≈ 1950 at large incidence angles. SPP mode in such a hexagonal hole array exhibits a sensitive linear monotonic dependence on the incidence angle. Additionally, a limit of detection concentration as low as 10^{-10} M for bovine serum albumin (BSA) has been achieved (Figure 5b).^[115]

Surface-supported metallic NP arrays suffer from the reduction of overall RI sensitivity due to the spatial occupation of the substrate.^[116] Elevated plasmonic nanostructures with dielectric pillars can expose a larger volume of hot spots to molecules than that of the nonelevated nanostructures for improving sensitivity.^[48,116] For example, sub-micrometer Au mushroom arrays lifted by photoresist pillars can support interference between Wood's anomaly and LSPRs to achieve a FOM of 108, which is comparable to the theoretically predicted upper limit for commercial SPP sensors (Figure 5c).^[35] These mushroom arrays were demonstrated to detect cytochrome c and alpha-fetoprotein with a low detection limit down to 200 pM and 15 ng ml⁻¹, respectively.^[35]

One-dimensional nanogroove arrays supported FP and WA modes can also be used for high-sensitivity sensing. Narrow resonances linewidth of 2 nm ($Q \sim 350$ in air and $Q \sim 400$ in water) and a FOM* of 617 under normal incidence in refractive index sensing is achieved from ultrasmooth Au nanogroove arrays.^[117] The high quality factor benefits from the geometrical-induced coupling between FP and WA modes in nanogroove arrays, which substantially suppress the out-of-plane radiation from individual nanogroove. And the surface scattering and grain boundary scattering can be minimized by template stripping to ensure a smooth surface (Figure 5d).^[117] Narrow resonances ($Q \sim 400$) with high experimental FOMs up to 284 in the RI range from 1.333 to 1.415 is demonstrated. Such a device can operate at normal incidence with a resonance intensity of $>50\%$ (Figure 5e).^[117] Further, the nanogroove array can also serve as a surface thickness sensor to detect mercaptocarboxylic acids with a surface sensitivity of 0.18 nm/CH₂, providing a promising platform for thickness detection of surface analytes and label-free biosensing (Figure 5f).^[117]

In addition, hybrid resonance mode such as quasi-bound states in the continuum (quasi-BIC),^[118,119] Fano resonance,^[120–122] and Tamm plasmon resonances^[123] enables the ultra-narrow linewidth to achieve high quality sensing. However, due to imperfect fabrication (e.g., grain boundary scattering, parasitic absorption of the adhesive layer, and inhomogeneous spectral broadening)^[124,125] and measurement limitations (e.g., inadequate collimation and finite spatial coherence of probing light source),^[126] devices with Q-factor of only ≈ 200 were achieved experimentally.^[127]

3.4. Plasmonic Nanogap Modes

Plasmonic nanogaps with sub-nanometer to a few nanometers can confine LSPRs and the associated electromagnetic field to small gaps. When the distance of nanogaps scales down to the sub-nanometer, a series of quantum effects, such as nonlocal electromagnetic effects, optical rectification, inelastic electron tunneling, and ring interference emerges.^[128] The promise of

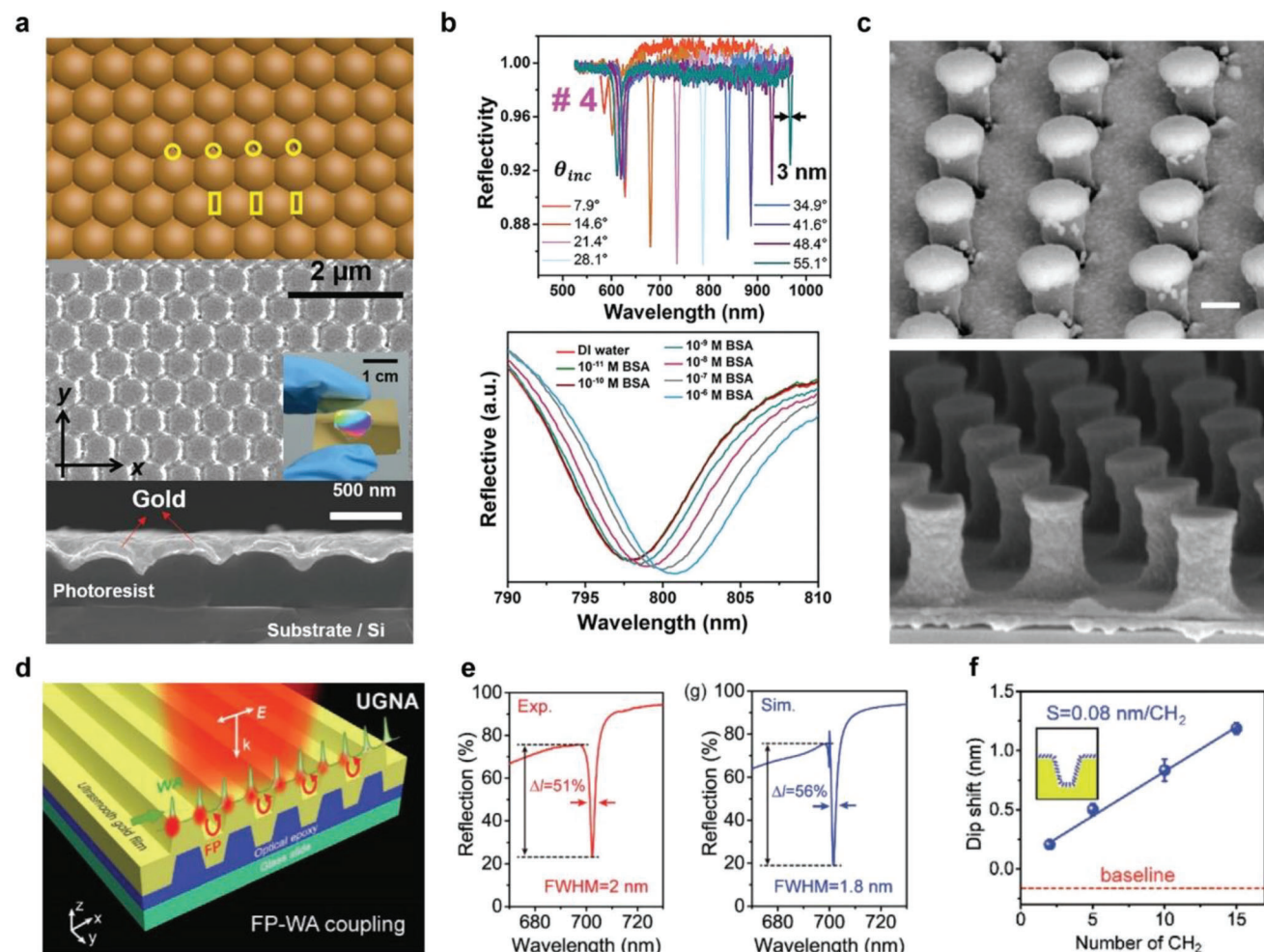


Figure 5. Patterned nanostructures with hybrid resonance modes for sensing. a) The schematic and SEM image of hexagonal Au arrays (periodicity = 620 nm and the thickness of Au film is 100 nm) supporting SPP modes. Nanoholes and nanoslits in arrays are highlighted with yellow circles and rectangles. b) Angle-dependent reflectance spectra of Au hole arrays showing the narrowest FWHM of 3 nm under an incident angle of 55.1° (top) and reflectance spectra of the sensor measured in solutions of different BSA concentrations (the bottom). c) SEM image of Au mushroom arrays. Top view and side view of photoresist arrays with Au caps (the thickness is 110 nm, periodicity = 610 nm, height = 510 nm). d) Schematic of 1D Au nanogroove arrays supporting FP-WA coupling mode. e) Experimental and simulation reflection spectrum of 1D Au nanogroove (periodicity = 700 nm, depth of the nanogroove = 345 nm) with an FWHM of 2 nm and 1.8 nm and resonance intensity of $>50\%$. f) 1D Au nanogroove arrays as surface thickness sensors for mercaptocarboxylic acids detection with dip shift as a function of the chain length of the analytes on the Au surface. (a-b) Reproduced with permission.^[115] Copyright 2018, Wiley-VCH. (c) Reproduced with permission.^[135] Copyright 2013, Springer Nature. (d-f) Reproduced with permission.^[117] Copyright 2021, Wiley-VCH.

nanogaps for sensing is extensively investigated when incorporated with other spectroscopy techniques. For example, nanogaps combined with Raman spectroscopy can identify molecules through their unique vibrational fingerprint.^[129] Tremendous works have been concentrated on the surface-enhanced Raman scattering (SERS) with nanogaps, such as plasmonic metal split rings with sub-nanometer split gaps,^[130] artificial plasmonic metal NP pairs,^[3] and 3D plasmonic metal nanostructures with sub-nanometer gaps.^[48] Suspended Au bowtie nanoantenna with sub-10 nm gap results in up to 2 orders of magnitude additional electromagnetic enhancement compared to that of nonelevated bowtie arrays.^[48] These suspended Au nanogap results in a SERS enhancement factor exceeding 10^{11} (Figure 6a). 3D star-shaped Au nanostructures with inter-particle spacing below 10 nm are

capable of few/single molecule detection by SERS (Figure 6b).^[45] Subsequently, bimetallic 3D nanostar dimers in ring cavities achieve recyclable and robust SERS substrates for signal detection of a few molecules (Figure 6c).^[131] Further, a 3 nm nanogap cavity enables surface-enhanced infrared absorption (SEIRA) with an enhanced factor of 10^5 due to the coupling of molecules and plasmonic nanogaps. Different resonance modes can be observed by tuning the cavity widths (Figure 6d).^[132]

Plasmonic nanojunctions composed of NPs on a mirrored geometry with a nanogap also exhibit potential for single molecule detection. For example, Au NPs are placed above a layer of Au film and separated with different conductivity self-assembled molecular spacer layers (Figure 7a).^[133] One atom difference between two molecules placed in the gap can result in a strong

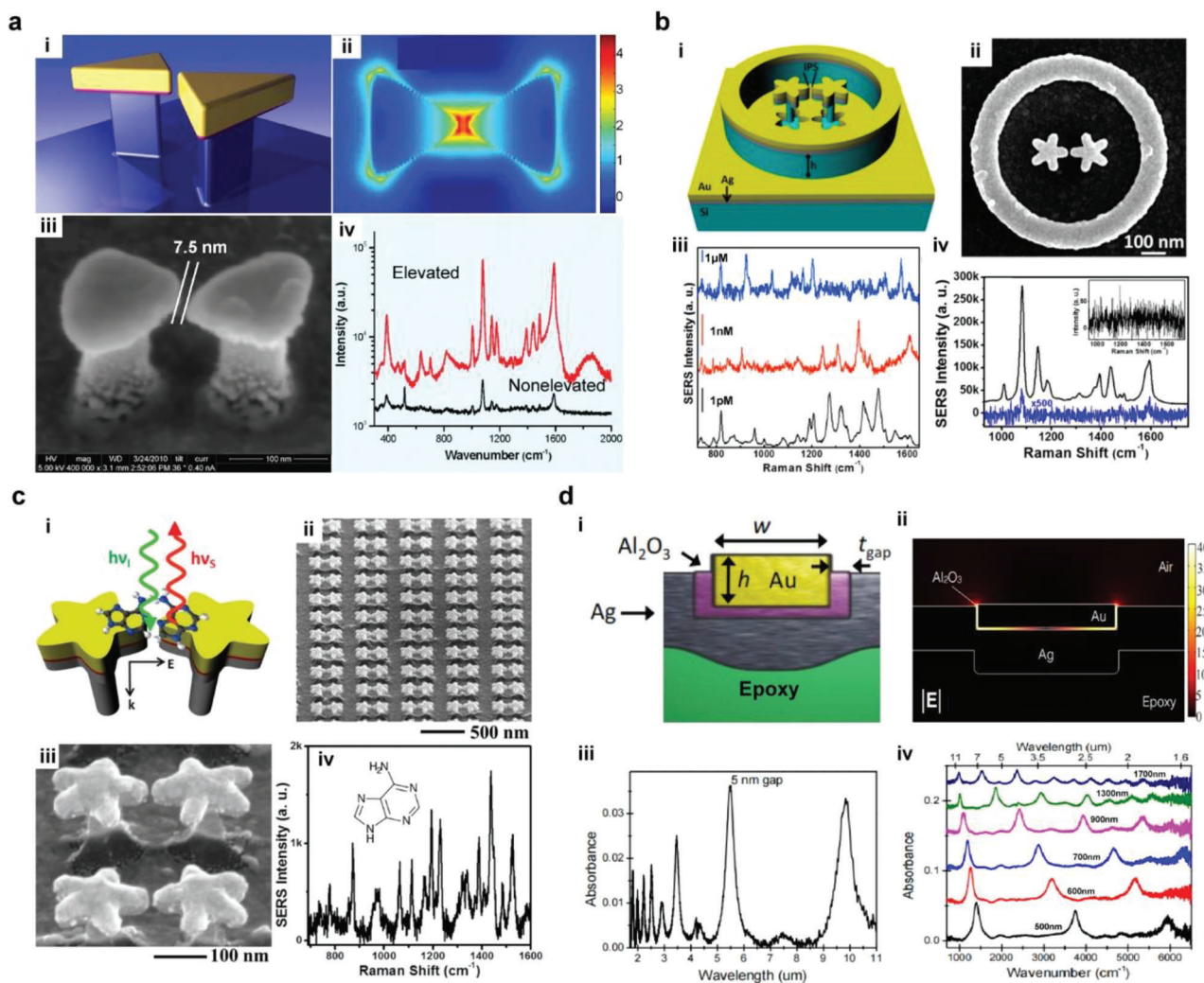


Figure 6. Plasmonic nanostructures arrays with nanogaps for sensing. a) i) Schematic, ii) near field distribution (triangle size of 100 nm, gap size of 10 nm, Si post of 40 nm in diameter, and 200 nm in height), iii) side-view SEM image of an Au bowtie unit with a gap of ≈ 1 nm, and iv) comparison SERS intensity of p-mercaptoaniline between elevated and nonelevated bowtie arrays. b) Bimetallic AgAu nanostar dimers with nanogaps below 10 nm in ring cavity arrays for sensing. i-ii) Schematic and top-view SEM image of nanostar dimers, iii) SERS intensities of adenine molecules at different concentrations, and iv) comparison of SERS spectrum of pATP on nanostar dimer arrays and Ag-Au film. c) i) Schematic for SERS principle and ii-iii) SEM images of nanostar dimers (the height of Si pillar is 150 nm), and iv) Raman spectrum with a solution at a 1-pM concentration of free-standing 3D Au nanostar arrays. $h\nu_i$ and $h\nu_s$ represent the incoming and scattering photon energy, respectively. The nanogap between nanostar dimers is below 10 nm. d) Nanogaps for SEIRA. i-ii) Schematic and near field distribution of the cavity with nanogaps (between Au and Ag) for SEIRA, iii) absorbance spectra of the cavity with a 5-nm nanogap, and iv) 3-nm nanogap cavities with different cavity widths. (a) Reproduced with permission.^[48] Copyright 2010, ACS. (b) Reproduced with permission.^[131] Copyright 2014, ACS. (c) Reproduced with permission.^[45] Copyright 2014, Wiley-VCH. (d) Reproduced with permission.^[132] Copyright 2014, ACS.

blue shift (≈ 50 nm) in the spectrum. By modeling the conductance of a single molecule and multiple molecules, the number of molecules in each plasmonic hot spot was quantified to be < 200 . Sensing sensitivity can be improved by decreasing the size of nanogaps, providing a reliable tool for single-molecule detection. An Au NP on a mirror spaced by the macrocyclic host molecule cucurbit[7] (CB[7]) achieves single-molecule SERS detection (Figure 7b).^[134] Further, quantum plasmonic immunoassay sensing is demonstrated in two nano plasmonic hemispheres cavity with a nanogap where hotspots reside.^[135] The detection limit can reach to single analyte due to the strong coupling between quantum emitter labels and plasmon polaritons with the

sensitivity increasing up to 1500% compared to the conventional shifting-type plasmonic sensors (Figure 7c). In addition, a similar structure was also used to probe the inhomogeneous field intensity distribution in the nanogap by employing a set of viologen-based, self-assembly monolayers as a molecular rule with ≈ 2 Å spatial resolution.^[136]

4. Hybrid Plasmonic Nanostructures

Plasmonic nanostructures based on metals have some intrinsic disadvantages. Most metallic plasmonic sensors have limited performance with nonadsorbing or weakly adsorbed molecules

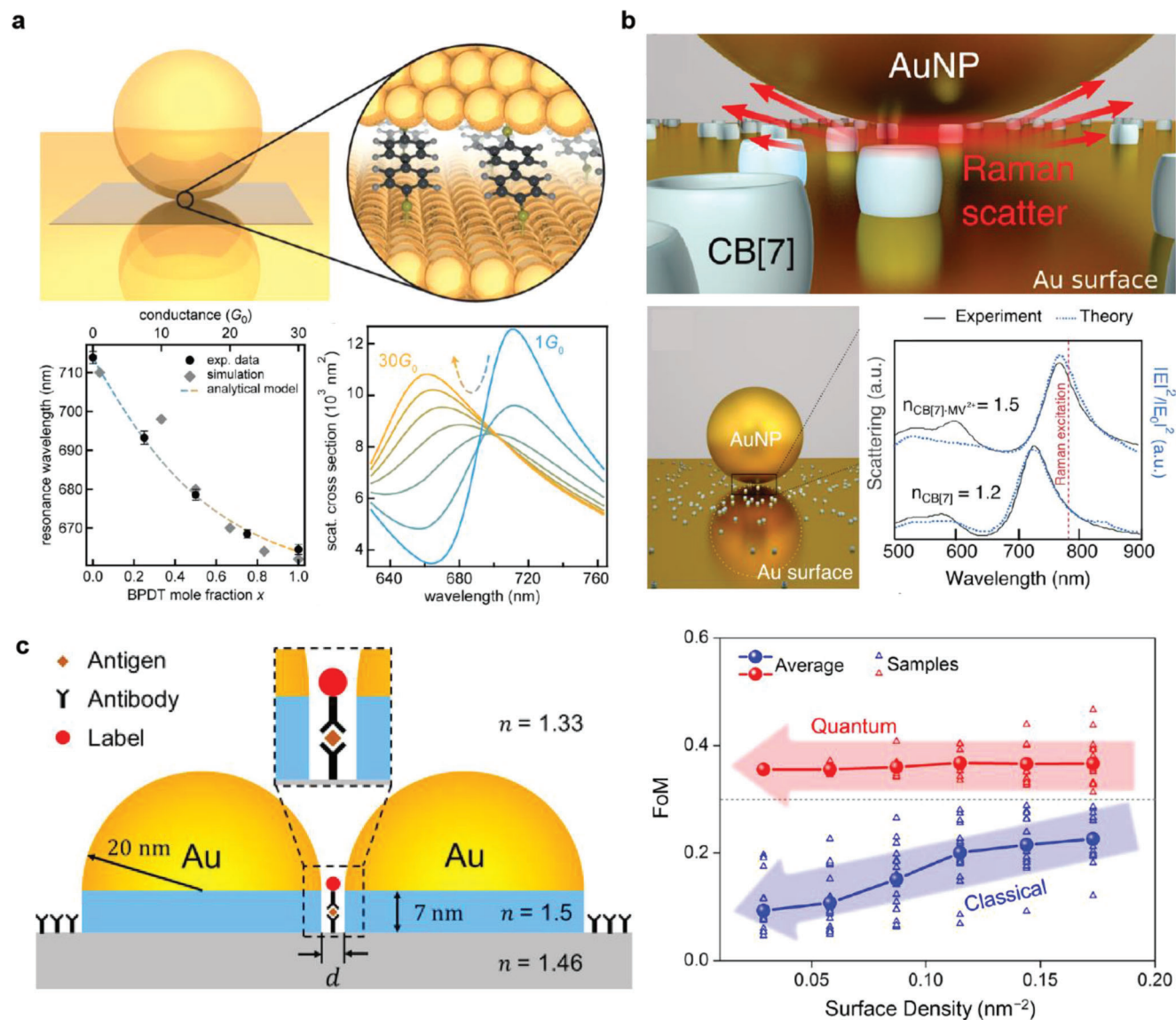


Figure 7. Plasmonic nanogaps for sensing. a) Schematic of an Au NP on the Au film separated with self-assembled monolayers (top) and conductance-dependent shift (bottom). b) Schematic of Au NP on the Au thin film spaced by CB[7] molecules and the scattering spectrum for SERS. c) Schematic of quantum immunoassay sensing with a nanogap (left) and comparison of FOM of classic and quantum structures (right). The analyte is captured in the plasmonic hotspots and resides in the nanogap of the Au hemisphere dimer cavity. (a) Reproduced with permission.^[133] Copyright 2015, ACS. (b) Reproduced with permission.^[134] Copyright 2016, ACS. (c) Reproduced with permission.^[135] Copyright 2019, ACS.

because of the weak interaction between them and the metallic surface.^[137] Additionally, bare metals, such as Ag, Cu, and Al NPs may suffer from cytotoxicity, poor chemical, and performance degradation in the aqueous and biological environment.^[42,138–140] Further, the Au NPs show good chemical stability but relatively poor thermal stability and high losses under irradiation.^[42,141,142] Al can extend plasmon resonances to the UV range with a d-band above its Fermi energy but usually support less strong resonances compared to Au and Ag.^[42,143] On the other hand, Pd is particularly suitable for optical hydrogen detection because of the index change after the formation of palladium hydride (PdH_x) upon hydrogen exposure.^[37,144,145] Sensing of bare Pd nanostructures tends to be deterred after the adsorption of poisonous gases, such

as CO and NO_2 .^[146] Therefore, hybrid nanostructures with functional materials, such as MOFs, oxides, 2D materials, and polymers were introduced to overcome the limitation of bare metals for sensing with high sensitivity and selectivity.

4.1. Plasmonic NPs with Metal-Organic Frameworks

Surface plasmons decay exponentially with distance, nonadsorbing target molecules are usually too far from the plasmonic surface or with low concentration. Surface adsorption becomes the limitation process for performance improvement of plasmonic sensing, especially for nonadsorbing molecules or weakly

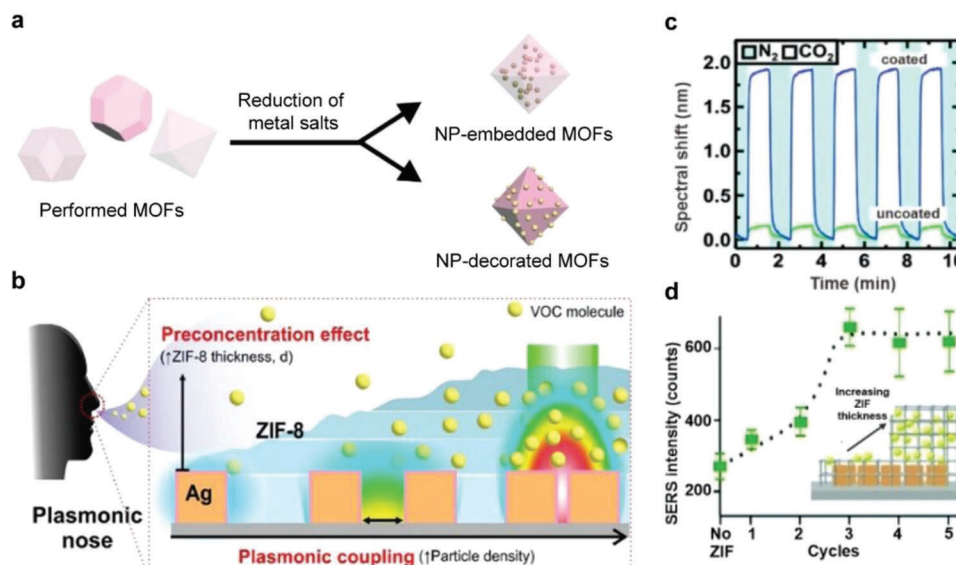


Figure 8. Sensing based on Plasmonic NPs with Metal-Organic Frameworks. a) Schematic of NP-MOFs nanostructures. b) Schematic of preconcentration effect of NP-MOFs nanostructures (Ag NPs coated with ZIF-8) platform for as plasmonic nose for volatile organic compounds (VOCs) sensing. c) Comparison of the spectral shift of the Ag LSPR sensor coated (blue line) and uncoated (green line) by HTUST-1 as a function of time. d) SERS intensity as a function of the thickness of ZIF. (a) Reproduced with permission.^[137] Copyright 2021, ACS. (b) and (d) Reproduced with permission.^[150] Copyright 2018, RSC. (c) Reproduced with permission.^[151] Copyright 2010, ACS.

adsorbed molecules such as toxic vapor, (e.g., benzene) and environmental toxins (e.g., organophosphorus pesticides).^[137] Gas detection at low concentrations down to ppb levels remains challenging for plasmonic sensors. Metal–organic frameworks incorporated with plasmonic NPs to form NP–MOF nanohybrids are proposed to detect nonadsorbing molecules at low concentrations. MOFs are three-dimensional porous crystalline materials composed of coordination of metal ions or clusters with organic linkers, which show ultrahigh-specific surface areas and molecular selectivity (Figure 8a).^[147,148] Plasmonic NP–MOF nanohybrids can accumulate analytes at the plasmonic nanostructure surface to increase the local concentration of analytes (Figure 8b).^[149,150] Tunable pore size in MOFs and different metal clusters or ligands can lead to a molecular sieving effect to improve the selectivity of analytes. For instance, HTUST-1 on Ag NPs acts as a molecular sponge and leads to a 14-fold enhancement in $\Delta\lambda_{\max}$ for CO₂ sensing compared to uncoated Ag NPs (Figure 8c).^[151] These hybrid sensors exhibit higher selectivity for CO₂ with thicker HTUST-1, which demonstrates that precise control of the MOF film thickness is critical.

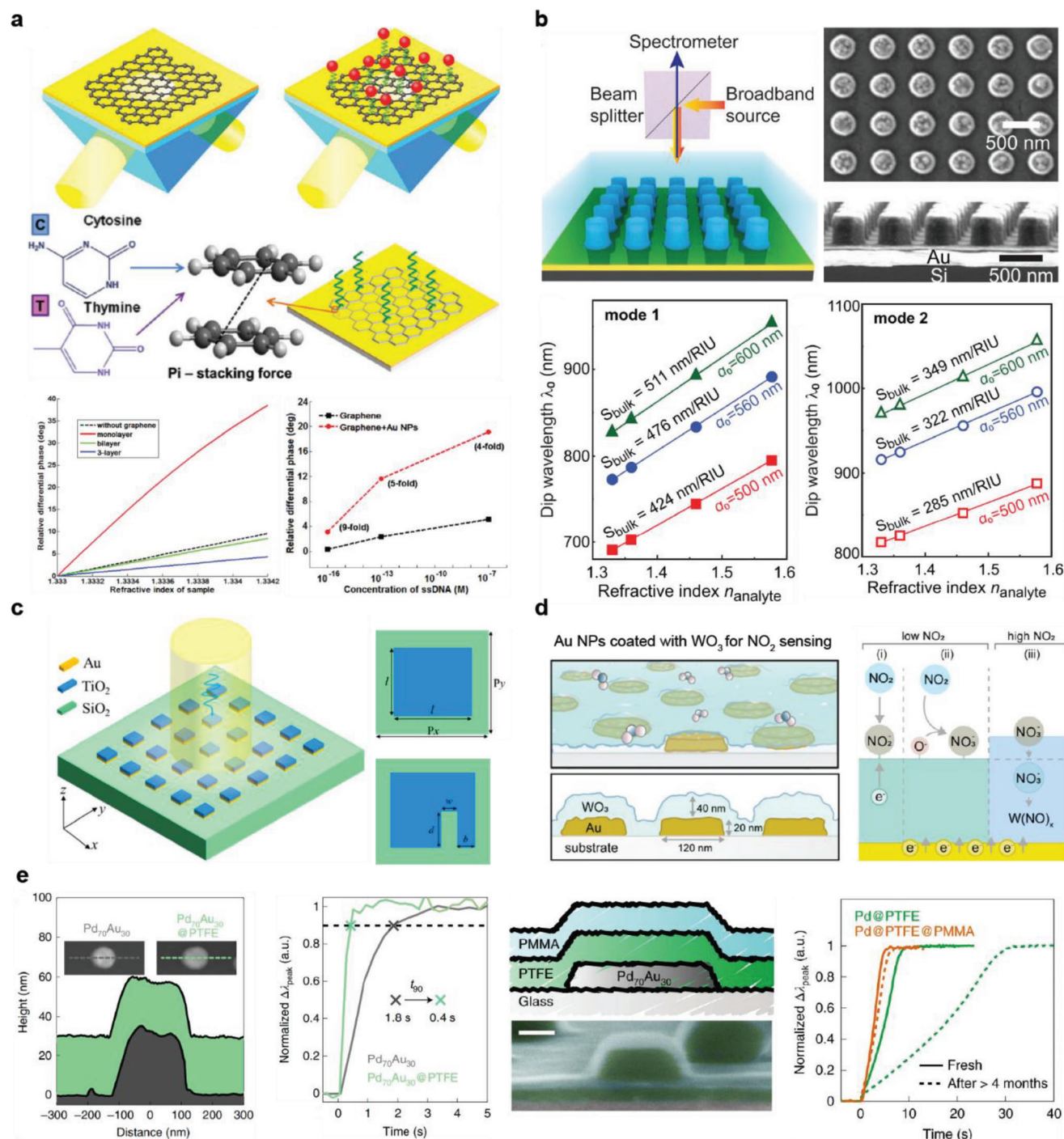
Combining MOFs with SERS structures can detect molecules with unique Raman vibrational footprints in part per billion (ppb) levels. Both the thickness of MOF coatings and the size of NPs are needed to be optimized to enhance the coupling and improve the SERS performance.^[152] Compared to bare Ag nanocube arrays, increasing the thickness of ZIF-8 from 8 to 146 nm over Ag nanocube arrays can achieve ≈ 2.5 -fold higher intensity and an analytical enhancement factor (AEF) of 7.6×10^6 (Figure 8d).^[150] Au cluster@ZIF-8 core–shell NPs can show a 10-ppb limit of detection for gaseous aldehydes.^[153] The thickness of the ZIF-8 can be used to precisely control the distance and electromagnetic fields between metal NPs, and a thickness of 3 nm shows good detection performance for the device volatile organic compounds

(VOC) gas molecules. Benefiting from the low cost, simple procedures, and rapid synthesis rate of MOF, this core-shell NPs base on MOF is promising for high-performance gas sensing.

4.2. Hybrid Metasurfaces with Functional Materials

Hybrid metasurface with functional materials, such as graphene and oxides can serve as high sensitivity and selectivity sensing with the possibility to achieve single molecular label-free detection. Hybrid graphene/Au metasurface offers a significant gain in phase sensitivity and selectivity to detect aromatic ring structure biomolecules through pi-stacking interactions (e.g., DNA, RNA, peptides, and cytokines) (Figure 9a).^[154–157] The field enhancement on the graphene sheet and Au-graphene interface result in extreme singularities in the phase of reflected light, leading to four times higher sensitivity compared to phase-sensitive SPR sensing.^[154] The response of phase to the change of RI exhibits proportional to the intensity of the probing field at the graphene–Au interface. These responses in Au-graphene metasurface exhibit four times stronger than the bare Au film, while two or more graphene layers do not enhance the response intensity.^[158] The detection limit of 10^{-18} M can be obtained in the detection of single-stranded DNA (ssDNA), which is lower than other reported plasmonic biosensors.^[154,159–162]

The intense optical field in the metal-based plasmonic nanostructures may result in dissipative loss and heating. Metal and dielectric hybrid metasurface can be an alternative to generate high electric field enhancement with low losses.^[163–166] TiO₂ nanopillar arrays on Au film were proposed for refractive sensing with a $Q > 200$ while the surrounding environment RI changes from 1.33 to 1.58 (Figure 9b).^[167] The hybrid structure displays sharp transitions in both intensity and phase of reflected light when



exposed to aqueous or organic environments, with a bulk sensitivity of 449 nm/RIU and a Q -factor of 300. Further, hybrid metasurfaces composed of Au and TiO_2 NP arrays can support Fano resonance from the destructive interaction between SPPs and LSPRs (Figure 9c).^[168] A single Fano resonance splits into multiple Fano resonances with a stub-shaped defect and can be tuned by the size and position of these defects (Figure 9d).^[168] Quality factor and dephasing time of the Fano resonance can be changed through the thickness of the Au and TiO_2 , unit spacing, and detection structures. The stub-shaped defect can enhance the sensitivity of these hybrid metasurfaces more than three times and reach 535 nm/RIU.^[168] NO_2 sensor based on arrays of Au NPs coated with a thin layer of polycrystalline WO_3 was tested in all idealized laboratory conditions, simulated humid urban air, and outdoor fields (Figure 9d).^[169] By taking advantage of the charge transfer between Au NPs and WO_3 as well as the volume expansion of WO_3 induced index modification, high sensitivity detection with LOD below 10 ppb was achieved.^[169]

Hybridization of polymers with plasmonic nanostructures to engineer plasmonic-analytes interfaces can improve the stability, sensitivity, selectivity, and response times of advanced sensors.^[144] Polymer components can enhance sensing performance by functioning as both media to improve surface transport and adsorption for higher sensitivity and absorption layers to selectively concentrate the target gas for higher selectivity.^[145,170] Optical detection of hydrogen gas requires ultrafast response time and sufficient resistance toward poisoning species, which can be achieved by coating a polymer layer on plasmonic nanostructures to improve hydrogen diffusion and block undesirable molecules. Pd or $\text{Pd}_{70}\text{Au}_{30}$ alloy NPs with a thin polytetrafluoroethylene (PTFE) cladding layer exhibit a subsecond response time for hydrogen detection (Figure 9e).^[146] The PTFE layer can reduce the apparent activation energy for hydrogen transport and suppress the hydrogen sorption hysteresis. This plasmonic-polymer hybrid shows an improved LOD (1 ppm in Ar and 5 ppm in synthetic air).^[146] The PTFE coating shows poor protection from deactivation by other gases, such as CO and NO_2 , while the poly (methylmethacrylate) (PMMA) cladding layer has excellent H_2 selectivity to other gases and protection for plasmonic NPs.^[171,172] $\text{Pd}_{70}\text{Au}_{30}$ @PTFE/PMMA tandem coating sensors were constructed to combine the properties of both PTFE and PMMA, and the device shows similar detection performance to that of PTFE coating alone and can operate in ambient conditions for four months without deactivation. Additionally, the polymer layer can also act as a medium to transfer the information of gas specie and concentration into RI changes through chemical or physical processes and followed by detecting through plasmonic responses. Selective chemical interactions between polymers and gases can lead to the polymer phase transition. For example, helix reversal can be formed after exposure to CO_2 in crystalline polymers and lead to rapid amplified RI changes.^[173] Combining plasmonic components with the phase transition PTFE substrate, selective CO_2 detection with relatively low sensitivity (LOD of 1000 ppm) is demonstrated.^[173] Even though performance improvements are needed, the mechanism of incorporating reaction-induced phase transition polymers with plasmonic nanostructures to enhance the selectivity is expected to be generic and provide promise for a wide range of sensor designs.

4.3. Core-Shell Nanostructures

Bare metal NPs for plasmonic sensing are unsuitable for many practical applications because of cytotoxicity, feature stability, and property degradations. For example, Ag NPs are easy to oxidize in the atmosphere, leading to attenuations of plasmonic properties and sensing performance.^[174] Au nanostructures are more stable than other plasmonic nanostructures but suffer from photothermal effects.^[175,176] Other undesired processes including aggregation, dissolution, and etching, can also degrade the properties of bare metal nanostructures for sensing.^[175] Therefore, inert thin-film materials such as silica, graphene, organic shells, and alloys have been utilized as coating layers to stabilize the noble metal NPs.^[174,175,177–182]

Au NPs with a silica protective layer can be employed to detect the hydrogen absorption process and reaction on the Pd film (Figure 10a).^[183] The silica shell serves as a spacer to separate Au NPs from direct contact with the probed reactant. When hydrogens diffuse into Pd film to form palladium hydride (PdH), the change of dielectric function of the Pd substrate results in a blue shift in the scattering spectrum (Figure 10b).^[184,185] Different concentrations of hydrogen ranging from 0.5% to 3% can be identified as intensity variations of the plasmonic resonance with response times on the order of seconds. Composite substrates consisting of Au nanodisks covered with a 5 nm silica layer could be equipped with a lipid bilayer to monitor the binding of a cholera toxin.^[186] Ag NPs have excellent plasmonic properties but are excluded from biosensing due to their cytotoxicity. Biocompatible silica layers can be coated on the Ag NPs to maintain the detection efficiency for bovine serum albumin.^[187] Silica coatings also play an essential role in quantitative SERS measurement. The hotspot usually resides at the sharp edge due to the lightning-rod effect or within strongly coupled particle clusters, hence the colloidal stability and reproducible interparticle spacing are important.^[30,188,189] Silica coatings allow the diffusion of analytes toward the plasmonic surface and preserve the metal NPs from uncontrolled aggregation to ensure strong and reproducible SERS enhancement.^[190,191]

Even with great promise, thick protection layers such as silica coatings may reduce the interaction strengths and attenuate the plasmonic properties in sensing applications.^[192] A single layer of graphene with a thickness of ≈ 0.35 nm is thinner but more impermeable than other coating materials.^[180,193] Graphene on Ag NPs effectively conserves the plasmonic performance and protects it from external contaminant sources such as hydrogen sulfide in the air.^[193] Compared with the bare Ag NPs, Ag@graphene NPs maintain complete morphology after one month of exposure to ambient air, indicating no sign of sulfidation and preservation of sensitivity (Figure 10c).^[180] Furthermore, graphene can also serve as an electron shuttle and deoxidize the Ag surface.^[193] Since the reduction potential of Ag oxides is higher than that of graphene, the injected electrons from graphene can result in the deoxidization of Ag oxides.^[194,195]

Bimetallic core-shell NPs can involve properties of both metals and shows a spectral character between them (Figure 9d).^[196–199] For example, Ag NPs support resonance ranging from 300 nm to 1200 nm while Au NPs can only support resonances with a wavelength longer than 600 nm due to inter-band transition, and Ag NPs exhibit higher intensities and stronger field enhancements

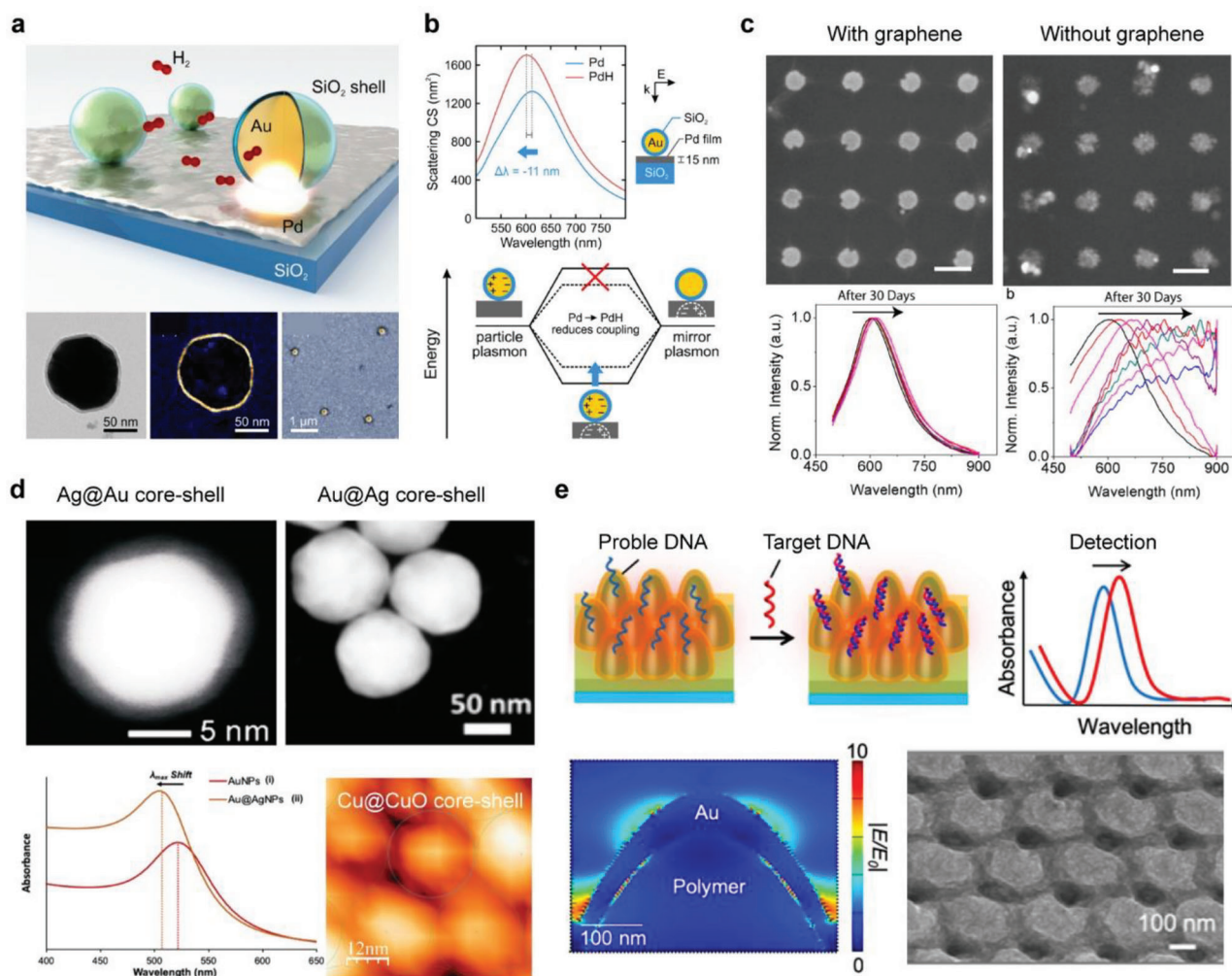


Figure 10. Core-shell nanostructures for plasmonic sensing. a) Schematic and SEM image of H₂ detection using Au@SiO₂ NP on a Pd film. b) Simulated scattering spectra before and after H₂ adsorption (top) and mode for interpreting the spectra blueshift. c) SEM image and normalized reflected spectra of Ag NP arrays coated and uncoated graphene layers. The coating of graphene significantly improves the stability of the NP arrays. d) TEM image of Ag@Au core-shell NPs, Au@Ag core-shell NPs, and AFM image of Cu @ CuO NPs and comparison of absorbance for Au NPs and Au@Ag NPs. e) Schematic of the DNA detection, near-field distribution, and SEM images of polymer@Au nanocone arrays. (a) Reproduced with permission.^[183] Copyright 2013, ACS (b) Reproduced with permission.^[183] Copyright 2013, ACS. (c) Reproduced with permission.^[180] Copyright 2012, ACS. (d) Reproduced with permission.^[197–200] Copyright 2017 ACS, 2012 IOP Publishing, 2020 ACS and 2011 ACS. (e) Reproduced with permission.^[201] Copyright 2019 ACS.

than Au NPs. Adding a small amount of Au (20%) in Ag as an alloy or core shell would significantly increase the absorption cross section due to interband transitions of Au.^[197] Cu@CuO core-shell NP array also enables a low flow rate of CO detection at room temperature with an average shift of 10^{-2} nm per molecule absorption.^[200]

In addition, nanocone arrays consisting of polymer core and Au shell show a refractive index sensitivity of 417.5 nm/RIU and can detect DNA with a detection limit of 161 fM (Figure 10e).^[201] The electric field enhancement and the excitation wavelength can be controlled by changing the thickness of Au shells.

5. Plasmonic Nanolasing for Sensing

Combining optical gain materials with plasmonic nanostructures can significantly suppress losses.^[202–206] Plasmonic nanolasing

that consists of a resonant cavity and gain medium manipulates and produces coherent light with ultranarrow linewidth and high intensity.^[207] Both the lasing wavelength and intensity have been employed for sensing applications. Losses can be compensated by the gain medium in nanolasing, which narrows the linewidth by two orders of magnitude and can dramatically improve the sensitivity of sensing.^[208] Further, nanolasers can also work as sensors based on the change of nanolasing intensity, whose sensitivity is much higher than the conventional surface plasmon resonance sensors because of their high spectral coherence and high signal-to-noise ratio.^[21,208,209]

By designing the nanocavity with a smaller mode volume and larger overlap between the optical field and analytes, high performance can be achieved for wavelength-based sensing.^[208,210,211] For example, a hybrid plasmonic crystal nanolaser featuring a ZnO nanowire (NWR) placed on Al grating with nano trench

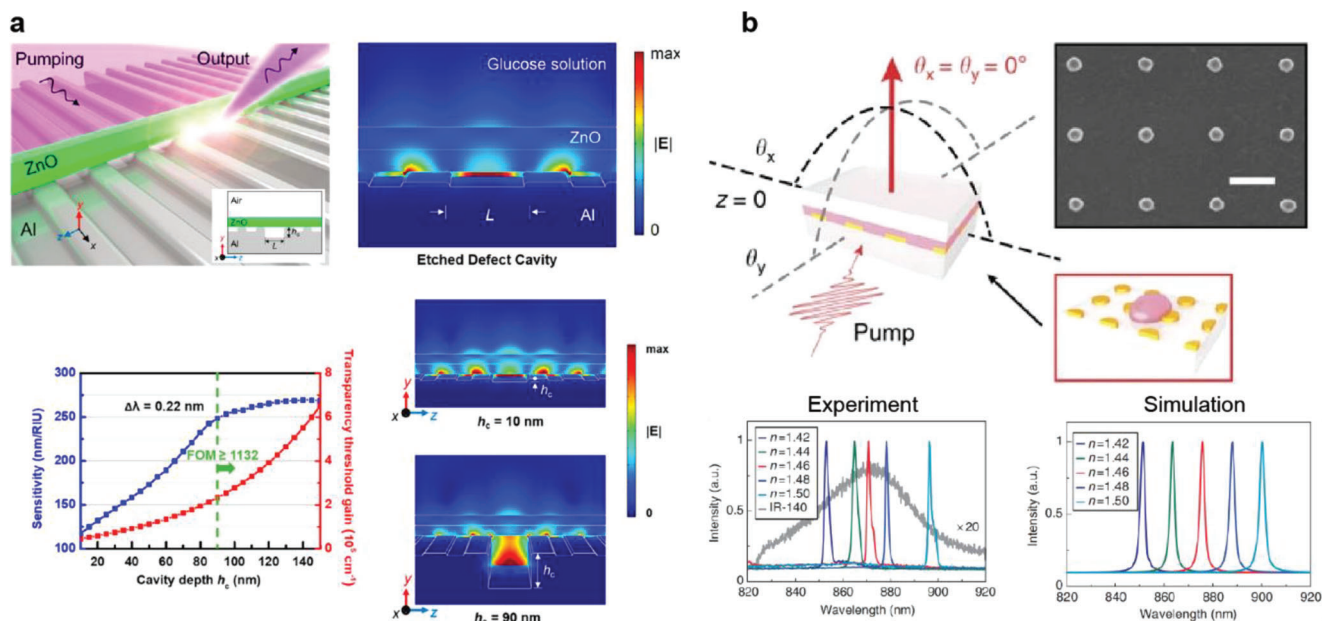


Figure 11. Nanolasing enhanced plasmonic refractive index sensing. a) Schematic of nanolasing from a ZnO nanowire on an Al substrate under optical pumping. Near-field distribution of ZnO nanowire nanolasing with etched defect cavity in glucose solutions, and their sensing sensitivity. b) Schematic, SEM image, and lasing spectra (both experiment and simulation) of nanolasing from Au NP arrays with dye solution. (a) Reproduced with permission.^[210] Copyright 2018, ACS. (b) Reproduced with permission.^[94] Copyright 2015, Springer Nature.

defect cavity can support mode volume as small as $5.75 \times 10^{-4} \lambda_0^3$ (Figure 11a).^[210] Compared to the NWR end facet without gratings, the Al gratings as a Bragg reflector reduce the mirror loss by 10-fold. This hollow nano trench defect between the Bragg reflector serves as a nanocavity with a high Purcell factor to enhance the interaction between light and matter (Figure 11a).^[210] Owing to the low loss and high Purcell factor, the sensing with the defect depth of 90 nm in glucose solution exhibits a high sensitivity of 249 nm/RIU with a FOM of 1132.^[210] This FOM is about one order of magnitude larger than that of previous plasmonic sensors and enables applications for real-time and label-free biosensing and gas detection in the near-UV regime.^[208]

Plasmonic lasing can sense the refractive index environment surrounding the device in real time, which offers the possibility for ultrasensitive RI sensing applications. The nanolaser consists of a two-dimension Au NP array and liquid gain media,^[94] liquid gain is created by dissolving the dye in different organic solvents as superstrate with different RI. The lasing signal can be tuned over the entire dye bandwidth with wavelengths ranging from 853 to 896 nm (Figure 11b).^[94]

Sensors based on plasmonic nanolasing can also monitor the lasing intensity variation and show higher sensitivity based on the index-induced wavelength change. For example, a sensor consisting of a single-crystalline semiconductor CdS nanoslab on top of the Ag surface, separated by a MgF_2 gap layer can detect explosives (e.g., 2,4-dinitrotoluene (DNT), ammonium nitrate and nitrobenzene) with a detection limit down to sub-ppb levels (Figure 12a).^[212] The surface plasmon localizes the electromagnetic field at the interface between the Ag surface and the CdS nanoslab, leading to both spatial and mode confinements in the nanometer scale perpendicular to the metal surface. The single-crystalline semiconductor CdS nanoslab with

an atomically smooth surface serves as optical gain materials, the geometry feedback, and the sensing medium simultaneously. The lasing intensity is tuned by the change of surface recombination velocity in the gain material that is sensitive to the adsorption of gas molecules. Monitoring variation of DNT concentrations is achieved by a continuous trace of lasing intensity (Figure 12b).^[212] The large surface-to-physical-volume ratio (S/V_{phy}) of the nanocavity, which is inversely proportional to the nanoslab thickness, ensures that any modulation of the surface will affect the emission intensity with the limitation of the number of carriers to be modulated inside. Therefore, any molecules that can be absorbed on the nanoslab surface would induce lasing intensity change, and identifying different molecules in a mixture remains challenging.

Nanolasing-based sensors in a biocompatible aqueous solution environment may degrade fast in their emission intensity due to the reaction of photogenerated holes with active sites at the crystal surface (Figure 12c).^[209] A 7-nm thickness Al_2O_3 layer as surface passivation on the CdS and CdSe gain materials can generate stable nanolasers to be over 68% in a biocompatible environment.^[209] Sensors with surface passivation can last 3600s without observable degradation, while the device without passivation can only last for a few seconds (Figure 12d).^[209] Serving as the refractive index sensor, the passivated nanolaser exhibits ≈ 8000 FOM for intensity detection, which is over 40 times higher than the state-of-the-art SPR sensors (Figure 12e).^[27,209]

To provide quick access to important information and key parameters of the plasmonic sensors covered in the previous text, we summarize the structural features and resonance modes of plasmonic nanostructures in Table 1. The corresponding parameters of performance evaluation for sensing applications are also included.

Table 1. Comparison of plasmonic nanostructures for sensing.

Material and structure	Resonance mode	Analytes	Figure of merit	Sensitivity	Q-factor	Incident angle	Limit of detection	Ref.
Au square array	SLR	Glycerol	120	–	85	0°	–	[101]
MIM array (Simulation)	SLR	–	–	–	147	3°	–	[107]
Pd NP array	SLR	Hydrogen	0.11	–	–	–	250 ppb (gas)	[37]
Au hole array	SPP	Bovine serum albumin	1944	5833 nm/RIU	–	55.1°	10 ^{–10} M	[115]
Au mushroom array	Interference between Wood's anomaly and LSPRs	Cytochrome c and alpha-fetoprotein	108	1015 nm/RIU	–	33.3°	200 pM	[35]
Au nanogroove array	Coupling of FP mode and Wood's anomaly	Mercaptocarboxylic acids	617	681 nm/RIU	400	0°	–	[117]
Au-graphene metasurface	SPP	DNA	–	–	–	52°	10 ^{–18} M	[154]
TiO ₂ NP array on Au film	SLR	–	180	449 nm/RIU	300	0°	–	[167]
TiO ₂ NP array on Au film	Fano resonance	–	91	535 nm/RIU	23.4	0°	–	[168]
Au NPs coated with a WO ₃ film	LSPR	NO ₂	–	–	–	–	About 3 ppb (gas)	[169]
Polymer@Au	LSPR	DNA	–	417.5 nm/RIU	–	–	161 fM	[201]
Pd ₇₀ Au ₃₀ NPs with PTFE coating	LSPR	Hydrogen	–	–	–	–	1 ppm (gas)	[146]
Nanolasing consists of ZnO nanowire on Al grating	SPR	Glucose solution	1132	249 nm/RIU	106	–	–	[210]
CdS nanoslab on MgF ₂ /Ag film	SPP nanolasing	DNT	–	–	–	–	Below 1 ppb (gas)	[212]
CdSe nanosquare with Al ₂ O ₃ layer on MgF ₂ /Ag film	SPP nanolasing	–	≈8000	–	–	–	–	[209]

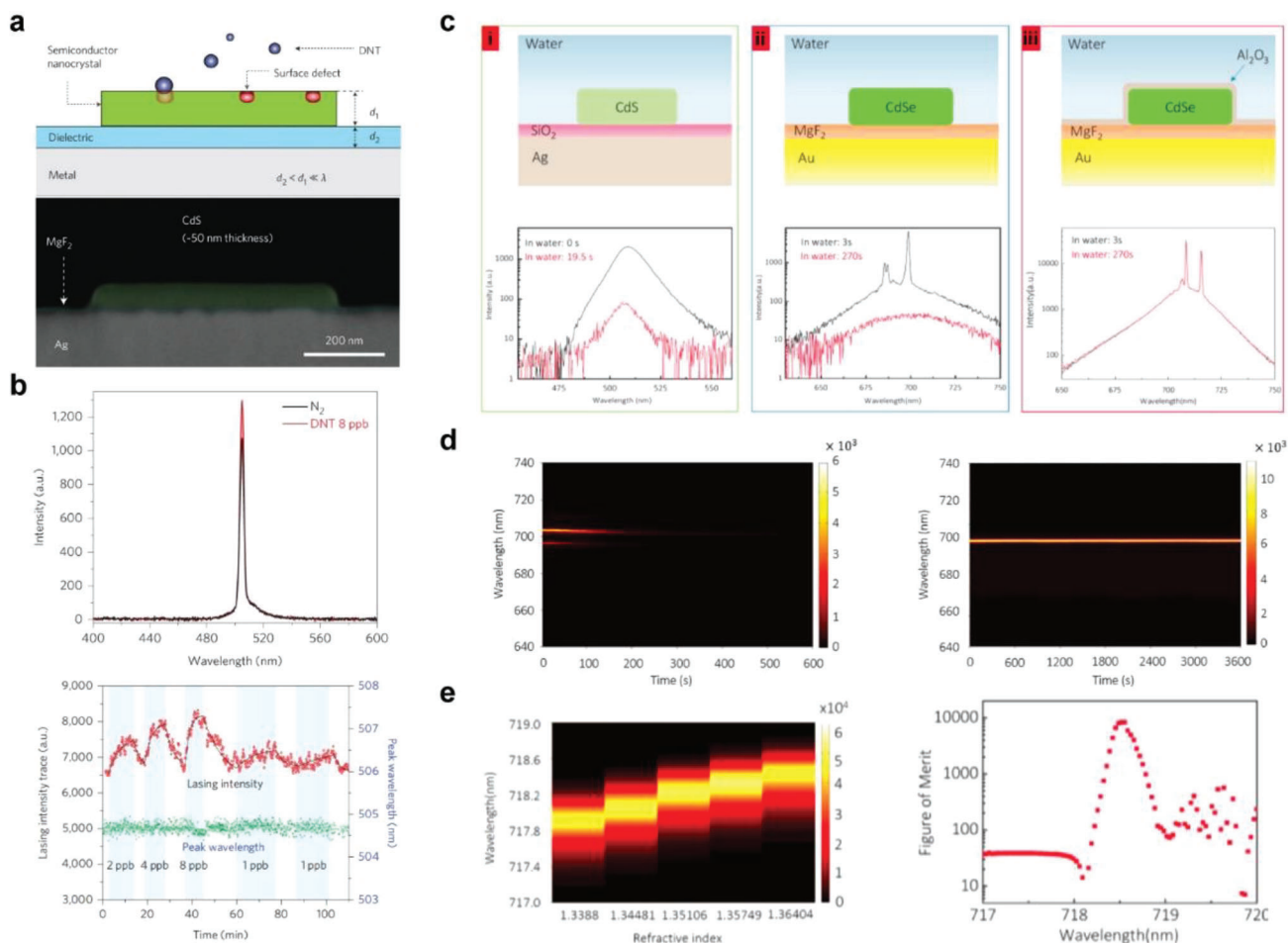


Figure 12. Plasmonic nanolasing for high-sensitivity sensing. a) Schematic (top) and SEM (bottom) of plasmonic nanolaser (CdS nanowire on MgF₂ film) for DNT detection. b) Experimental spectra of nanolasing with N₂ and 8-ppb DNT (top), the trace of continuous nanolasing intensities at different DNT concentrations (bottom). c) Schematic and traces of three different structures (top) and a trace of lasing intensities (bottom) in water. i) CdS nanosquare on the SiO₂/Ag substrate shows no lasing in water. ii) CdSe nanosquare on the MgF₂/Au substrate can lase but only can last for tens of seconds (without protection). iii) CdSe nanosquare on the MgF₂/Au substrate with Al₂O₃ protection cladding layer can lase stably in water. d) Comparison of a continuous trace of emission intensity of CdSe nanosquare without (left) and with the Al₂O₃ protection layer (right). e) Lasing spectra of CdSe nanosquare with Al₂O₃ protection layer under five RIs (left) and its FOM as a function of the wavelength (right). (a-b) Reproduced with permission.^[121] Copyright 2014, Springer Nature. (c-e) Reproduced with permission.^[209] Copyright 2017, ACS.

6. Plasmonic Materials Beyond Noble Metals

Most plasmonic nanostructures are made from noble metals due to their strong optical response over visible and near-infrared wavelengths. However, noble metals suffer from large losses mainly arising from interband electronic transitions, especially in the visible and UV spectral ranges.^[42,213,214,142] These losses result in broad linewidths in the spectrum, which severely hinders the sensitivity of plasmonic sensing. Combining optical gain materials to offset the metallic losses is proposed as a solution,^[202–206] however, even the best gain materials are barely enough to compensate for the inherent losses of the materials.^[142] Hence, alternative materials with lower losses are necessary for high-performance plasmonic sensing. Various materials, such as alkali metals, metallic alloys, highly doped semiconductors, and 2D materials are potential candidates in terms of their relevant properties for plasmonic sensing.^[142]

6.1. Metals Beyond Noble Metals

Alkali metals with lower intraband damping rates than noble metals can be promising candidates for plasmonic platforms.^[215,216] Among the alkali metals, Na and K exhibit higher FOM than Ag with the lowest losses.^[40,142,215,217,218] Na-based plasmonic waveguides and plasmonic nanolasers in the NIR range have been achieved.^[218] The Na-based nanolaser exhibits a single-mode lasing with a low threshold of 140 kW cm⁻². The Na film can be fabricated by the thermal-assisted spin-coating process in a glove box with an inert atmosphere.^[218] By coating with quartz and epoxy, these sodium-based devices can be stable over a long period.^[218] Na nanopit arrays with varying periodicities (300–600 nm) and tunable SPP modes (resonance linewidth down to 9.3 nm) in the visible to NIR range were recently demonstrated (Figure 13a).^[219] However, the high chemical reactivity of the alkali metals limits their properties for

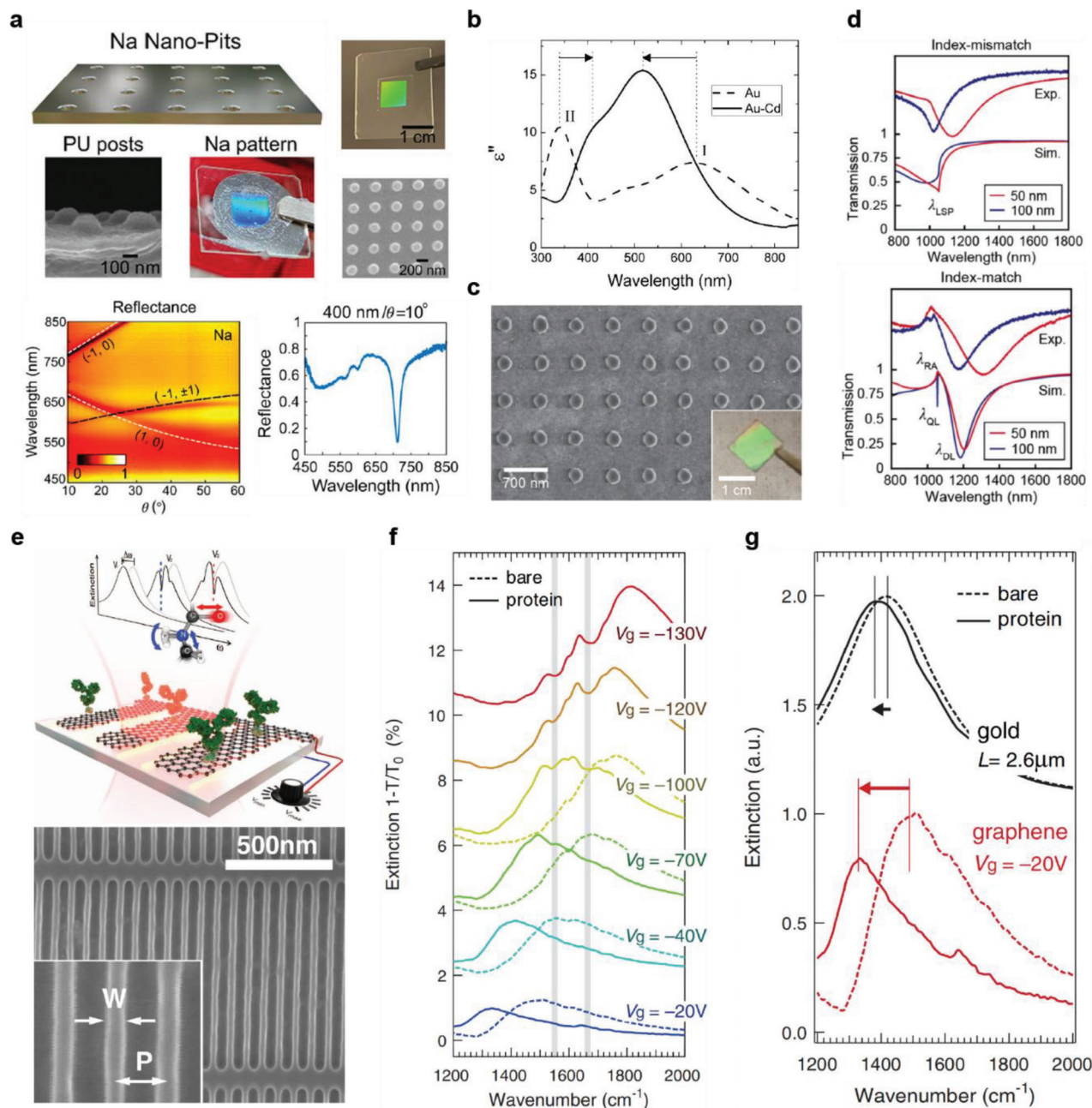


Figure 13. Applications of plasmonic materials beyond noble metals. a) Schematic and nanostructure characterization of the Na nanopit arrays (top) and dispersion diagrams and an example reflectance spectrum of Na nanopit arrays (bottom). b) The shift of interband transition peak of Au doped with 3.3 at. % Cd. c) SEM image and photo of TiN NP arrays. d) Experimental and simulated transmission spectra of TiN NP arrays under index-mismatch and index-match conditions. e) Schematic (top) and SEM (bottom) of tunable graphene mid-IR biosensing. The width of the graphene nanoribbon is 30 nm and the periodicity is 80 nm. f) Experimental extinction spectra of graphene nanorod arrays for protein detection with different bias voltage. g) Comparison extinction spectra of Au nanoantenna arrays and graphene nanorod for protein detection. The size of Au nanoantenna is $2.6 \mu\text{m} \times 0.2 \mu\text{m} \times 0.1 \mu\text{m}$ and the bias voltage of graphene nanoribbon is -20 V . The sensing sensitivity of the graphene nanorod exhibits ≈ 6 times higher than that of the Au nanostructured sensor. (a) Reproduced with permission.^[219] Copyright 2023, ACS. (b) Reproduced with permission.^[42] Copyright 2010, Wiley-VCH. (c-d) Reproduced with permission.^[223] Copyright 2021, ACS. (e-g) Reproduced with permission.^[224] Copyright 2015, AAAS.

practical applications since they need to be stored in mineral oil or ultrahigh vacuum environments. Further studies for alkali-based plasmonic sensors with safe and easy fabrication procedures are still needed.

Metallic alloys, intermetallics, and mixed metallic compounds with large free electron densities are candidates for alternative plasmonic materials.^[42] Alloying two or more noble metals together can achieve a tunable band structure by adjusting the

proportion of each component.^[42] For example, the alloys of Au and Ag with slightly improved properties are promising materials in the visible wavelength range. The n-type doping would shift the threshold of interband transitions, thereby modifying the optical spectra of the alloy. It has been theoretically demonstrated that changing the Fermi level of metallic layers can tune the absorption at a specific wavelength.^[42,220] Desirable electronic properties can be achieved by tuning the electronic band structures of alloys. When doping the bivalent transition metals (e.g., Cd) into Au, the Fermi level of Au would be raised by contributing two electrons to the free-electron plasma.^[220] The peak from interband transition losses can be shifted to a new frequency (Figure 13b).^[220] Therefore, the high loss can be confined to a region, and losses at other frequencies can be reduced. However, this method is only suitable for low doping concentrations because metal structures would break down as the doping levels become too high.^[42] Alkali-noble inter-metallic compounds with strong free-electron-like behavior are also proposed as potential candidates for low-loss plasmonic materials.^[221,222] However, the fabrication of the alkali-noble inter-metallic compounds remains challenging due to the wide difference in vapor pressures and surface energies between the alkali and noble metals.^[42] Further, the phase, stoichiometry, and growth kinetics of these alloy and intermetallic systems still need further study.

6.2. Highly Doped Semiconductors

Highly doped semiconductors with high mobility, negative real permittivity, and nearly no interband transition losses in the visible range due to wide bandgaps, are promising alternative plasmonic materials.^[225–228] Indium-tin-oxide (ITO), a transparent and conducting oxide typically consisting of 90 wt.% indium oxide (In_2O_3) and 10 wt.% tin oxide (SnO_2), is considered as a potential plasmonic material in the NIR region.^[229–232] The plasma frequency can be tuned by varying the Sn doping level.^[233] Plasmon frequencies at 1.8 μm and 1.9 μm in SPPs by ITO have been experimentally achieved.^[234] The optical properties of ITO mainly depend on the growth/deposition processes and annealing conditions, including the temperature and ambient gasses.^[226] High conductivity of the ITO film can be realized in a high annealing temperature for lower loss within the wavelength range of interest,^[42] for instance, ITO films show comparable loss to Ag within the NIR range and lower at a longer wavelength. High-conductivity zinc oxides such as aluminum-zinc-oxide (AZO) and gallium-zinc oxide (GZO) are also promising low-loss plasmonic materials.^[42,235,236] AZO exhibits losses more than three times lower than Ag at 1.5 μm .^[235,237,238] Temperature and oxygen partial pressure during deposition and dopant concentration can be optimized to obtain low loss and negative real permittivity in AZO and GZO films.^[235–239] Importantly, high doping in AZO cannot effectively increase the carrier concentration because of the formation of ZnAl_2O_4 , but reduce the carrier mobility with increased scattering.^[240]

In addition to oxides, heavily doped III-V semiconductors (e.g., GaN, GaAs, GaP, etc.), SiC, and TiN with wide bandgaps exhibit prominent optical properties for alternative plasmonic materials in NIR and visible spectral ranges.^[223] TiN NP arrays have been demonstrated to remain intact at pump power 2–3 orders of mag-

nitude higher than the saturation fluence of Au NPs (Figure 13c). Both dipolar SLR and quadrupolar SLR can be supported in TiN NP arrays in the NIR range (Figure 13d).^[223] Additionally, the possibility of growing TiN on Si (100) may lead to the integration of plasmonics with nanoelectronics.^[217,241,242] For intrinsic semiconductors, high doping concentration can transform the crossover frequency into optical frequency. For example, GaAs with doping of $7 \times 10^{18} \text{ cm}^{-3}$ raise the crossover frequency to $\approx 9 \mu\text{m}$,^[228] but a higher doping concentration of at least $3 \times 10^{20} \text{ cm}^{-3}$ is required for optical frequency. Achieving such high-concentration doping remains challenging due to the limits of solid solubility and doping compensation effects.^[243,244] Maintaining high carrier mobility under high doping levels is difficult because of scattering but essential for low losses applications.

6.3. 2D Materials

Graphene with high carrier mobility and zero band gap is promising for plasmonic applications.^[226,245–249] Surface plasmons can be supported in graphene with a dispersion relation different from plasmon in 3D materials.^[250,251] The advantages of graphene for alternative plasmonic material can be summarized into four aspects. Firstly, doped graphene exhibits lower loss in various loss mechanisms.^[252] The interband transition loss that constitutes the major loss can be tuned by different doping levels.^[253,254] Secondly, graphene enables larger spatial light confinement than metals, which benefits light-matter interactions and integrated IR photonics.^[255–257] Thirdly, the mid-infrared (mid-IR) response of graphene is suitable for the detection of biomolecules, such as proteins, lipids, and DNA. Finally, the long-lived collective electron oscillations in graphene can be flexibly tuned by external bias. Based on these outstanding attributes, a high-sensitivity and tunable mid-IR biosensor based on graphene has been proposed for protein detection.^[224] The device consists of a layer of graphene nanoribbon array and a 280-nm-thick SiO_2/Si substrate (Figure 12e).^[224] The Fermi level of graphene and the peak position in the spectrum can be tuned by bias voltages to cover the amide I and amide II bands for protein detection (Figure 13f).^[224] Under similar conditions, graphene exhibits a resonance shift of 160 cm^{-1} after protein bilayer formation, which is ≈ 6 times higher sensitivity than that of Au nanostructured sensors (Figure 13g).^[224]

Transition metal dichalcogenides (TMDs, e.g. WS_2 , MoS_2) with tunable bandgaps and high absorption coefficient in visible and NIR light show potential in plasmonic sensing.^[258–260] Because of stronger optical absorption in monolayer TMDs ($\approx 5\%$ – 10%) than graphene ($\approx 2.3\%$), TMDs exhibit higher energy transformation to metals and larger electric field enhancement.^[261–263] Hybrid graphene-TMDs and metal-TMDs configurations based on SPP sensors have been proposed, for example, Au- MoS_2 -graphene structures were predicted to have more than 500-fold enhancement compared to Au-graphene.^[264,265] Theoretically, the sensitivity increases with the increment of TMDs layers number owing to stronger light absorption and would reach a maximum at a certain number of layers. Notably, most studies on TMD based on plasmonic sensors are theoretical calculations, and more experimental demonstrations and evaluations are needed.

7. Summary and outlook

Despite the great promise of plasmonic nanostructures for high-sensitive sensing, the development of new theories and technologies is still desirable, including new resonance modes, low-loss plasmonic materials, chirality sensing, novel plasmonic device structures, and fabrication of scalable high quality plasmonic nanostructures.

7.1. New Resonance modes from Low-Loss Materials

Plasmonic materials with versatile and remarkable properties have led to new fundamental insights and numerous applications because they have been almost the sole choice to achieve efficient control of light at the subwavelength scales. Compared to plasmonic materials, high refractive index dielectric materials can intrinsically support both electric and magnetic Mie resonances with substantially lower ohmic losses (Figure 14a).^[266,267] Nanophotonic devices of high-index all-dielectric nanostructures have emerged as another platform to explore subwavelength optics and photonics.^[266,268,269] With a random lattice structure, the all-dielectric sample shows moderate sensitivity to the RI changes with a sensitivity of five to ten times lower than their plasmonic counterparts. The lattice arrangement of the dielectric resonators to form metasurfaces can significantly enhance the sensing performance by supporting high-Q resonances.^[267] For example, metasurfaces with the Si unit consisting of a nanobar and a nanodisk can couple the electric dipole mode of the nanobar resonator and magnetic dipole mode of the nanodisk resonator to support Fano resonances at the telecommunication wavelength (Figure 14b).^[270] Anapole resonance from destructive interference between the electric and toroidal dipoles has also been used to generate narrow resonances in the visible regime for high-sensitive sensing (Figure 14c).^[271,272] Furthermore, dielectric structures can support quasi-bound states in the continuum (quasi-BICs) by mode coupling and destructive interference between several Mie resonances with a similar far-field pattern (Figure 14d).^[273] The radiative loss can be dramatically suppressed and high quality factor up to 18500 has been shown in asymmetric dielectric metasurface supporting quasi-BIC modes (Figure 14e).^[274] We anticipate these new resonance modes in dielectric structures with such high quality factors can contribute to a new regime for sensing applications.

7.2. Chirality Sensing from Plasmonic Nanostructures

Chirality is ubiquitous in nature biomolecules such as proteins, amino acids, and carbohydrates. Chiral sensing and detection are critical for molecule interactions and biomedical applications in the chemistry, biomedicine, food, and pharmaceutical industries.^[275,276] Circular dichroism (CD) is a widely used and reliable method for the analysis of biomolecules, but CD signals are usually weak due to low concentration and small characteristic size. Chiral molecules can interact with plasmonic nanostructures to show stronger CD based on two different mechanisms. First, strong near-field enhancements of plasmonic nanostructures and metasurfaces with achiral features have been proposed

to enhance the CD signals by engineering their coupling with optical resonances (Figure 15a).^[277–280] Plasmonic enhancements of chiral signals in Raman scattering under circularly polarized lights were also demonstrated.^[281] Further, chiral sensing based on plasmonic nanostructures and metasurfaces with chiral features is also attractive. Chiral responses can be generated either from individual chiral resonators or chiral-arranged structures composed of achiral units which can be prepared by both bottom-up synthesis approaches and top-down lithography methods (Figure 15b).^[282–287] Spiral and helicoid structures are often designed to support the circulation of electric charges to induce enhanced magnetic near fields for chiral responses. For example, nanohelices can detect modifications of RI with high sensitivities ($(1,091 \text{ nm RIU}^{-1})$ at $\lambda = 921 \text{ nm}$) and FOMs ($>2,800 \text{ RIU}^{-1}$) based on the changes of amplitudes and positions of the maxima as well as positions of the zero-crossing points in the CD spectra (Figure 15c).^[288] Moreover, chiral NPs synthesized using colloidal chemistry can be used as platforms to generate superchiral near fields. Importantly, local hotspots in conventional plasmonic chiral nanostructures are limited by their localized features that tend to be affected by the target molecular orientations, vibrations, and local concentrations.^[282,289] To solve this issue, collective CD from arrays of plasmonic chiral NPs were developed to significantly enhance enantioselective sensing (LOD of 114 pM) by supporting a strong and uniform chiral near fields over a large volume above the array plane.^[290] Each NP's electric dipole is like a spinning top and spins in the same direction at the same speed under the CPL, which produces a uniformly chiral electromagnetic field across the array (Figure 15d). Besides strong electric resonances in plasmonic structures, dielectric nanoantennae and their metasurfaces can generate both electric and magnetic multipoles that exhibit strong chirality. Hybrid structures of plasmonic and dielectric materials may be promising candidates to support superchiral near fields for chiral sensing, while plasmonic parts can maintain strong near field enhancement and dielectric parts can reduce optical losses.

7.3. Device Innovation of Plasmonic Sensors

Besides pursuing stronger near-field enhancement and narrow resonances as mentioned in previous parts, engineering sensor devices could be an effective strategy to further improve the detection performance. For example, diffractively coupled plasmonic NP arrays under oblique incidence exhibiting topologically protected zero reflection with extremely fast phase changes can radically improve the sensing sensitivity.^[291] Further, a pixelated dielectric metasurface with patch arrays of resonator lattices supporting a range of ultrasharp resonances was demonstrated for high-performance sensing. This superlattice-like metasurface enables molecular recognition and fingerprint information readout from direct extraction of the high quality total reflectance signals and a barcode-like spatial absorption map (Figure 16a).^[292] Similar to the sensing devices based on these superlattice arrays, sensors with ultrasharp discrete resonances can also be constructed from a single metasurface with tunable light incidence angles and polarizations (Figure 16b). In another example, Al trench arrays can concentrate molecules near hot spots during analyte solution droplet evaporation, which significantly

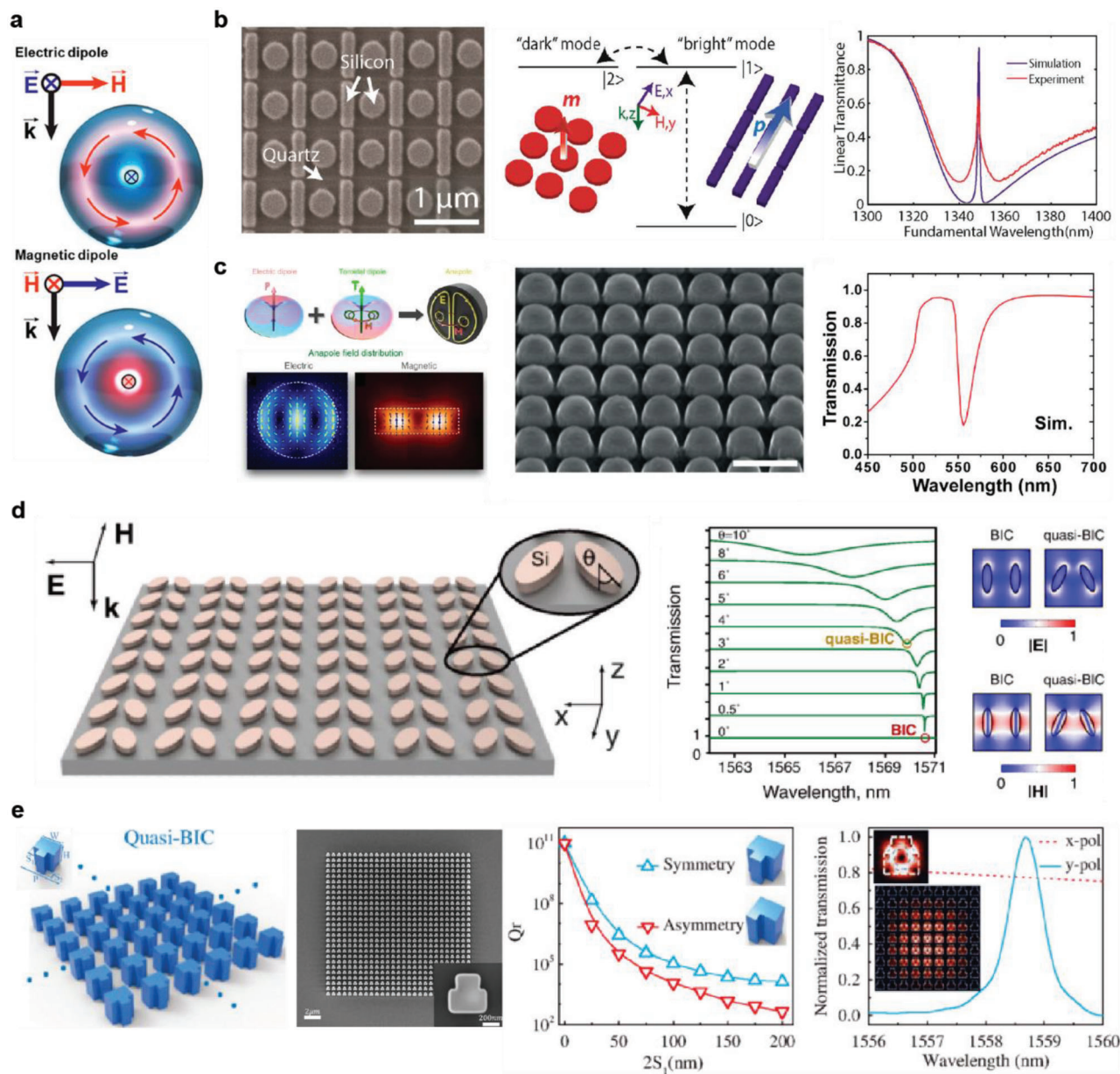
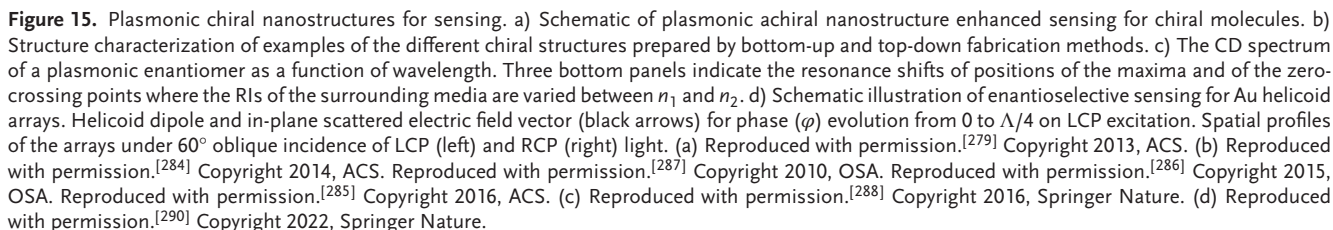


Figure 14. New resonance modes from low-loss materials for sensing. a) Schematics of an electric dipole and a magnetic dipole induced in a spherical dielectric NP. The red and blue arrows indicate the direction of the induced magnetic and electric field inside the NP, respectively. b) The SEM image of Si metasurfaces that can support Fano resonances and schematic of the Fano interference between the bright and dark modes. Simulated (blue) and experimentally measured (red) transmittance spectra of the Si metasurface. c) Schematic of an anapole excitation and electric and magnetic scattered near-field distribution at the anapole wavelength. The SEM image and simulated transmission spectra of a TiO_2 metasurface that can support anapole modes. d) Schematic of quasi-BIC metasurfaces with tilted Si ellipses. Transmission spectra versus the rotational angle θ . Simulated distribution of the electric and magnetic fields for both BIC and quasi-BIC. e) Schematic and SEM images of Si metasurfaces with symmetric defects that support quasi-BICs. The radiative quality factor of quasi-BIC eigenmodes (infinite size) versus the degrees of in-plane symmetry breaking for two types of symmetry-breaking methods. The transmission spectra of 9×9 Si blocks with symmetric defects. (a) Reproduced with permission.^[267] Copyright 2020, ACS. (b) Reproduced with permission.^[270] Copyright 2015, ACS. (c) Reproduced with permission.^[271] Copyright 2015, Springer Nature. Reproduced with permission.^[272] Copyright 2019, ACS. (d) Reproduced with permission.^[273] Copyright 2018, APS. (e) Reproduced with permission.^[274] Copyright 2019, APS.

enhances the spectral response and sensing performance (Figure 16c).^[293] Electric fields have been used to align the orientation of antibodies to enhance the binding probability between the antibody and antigen, thus performances of plasmonic sensors are improved with bias during antibody immobilization.^[294]

Further, parity-time (PT) symmetry refers to a system that commutes under combined parity and time-reversal operations with a balanced gain and loss profile which has been demonstrated within different optical systems such as two coupled microdisks and photonic crystal parallel waveguide structures.^[295]



lattice arrangement or nonidentical NP units. Advances in the fabrication of metal NPs lattices with single-crystalline, uniform shapes, smooth, and stable NPs are highly desirable. Thermal annealing and chemical vapor deposition procedures have been developed to achieve ultranarrow SLRs with full width at half-maxima linewidths toward the theoretical limit from arrays of Au, Ag, Al, and Cu NPs in a recent study (Figure 16d,e).^[298,299] The potential of this annealing posttreatment method is expected to be further expanded with the development of cost-effective methods to generate a large-scale ordered pattern of plasmonic nanostructures. Nanoimprint, phase shift lithography, and projection reprogramming may serve as good candidates to realize wafer-scale patterns of NPs with nanoscale precision.

7.5. Summary

In summary, plasmonic structures can confine the light below the diffraction limit and provide a reliable platform for

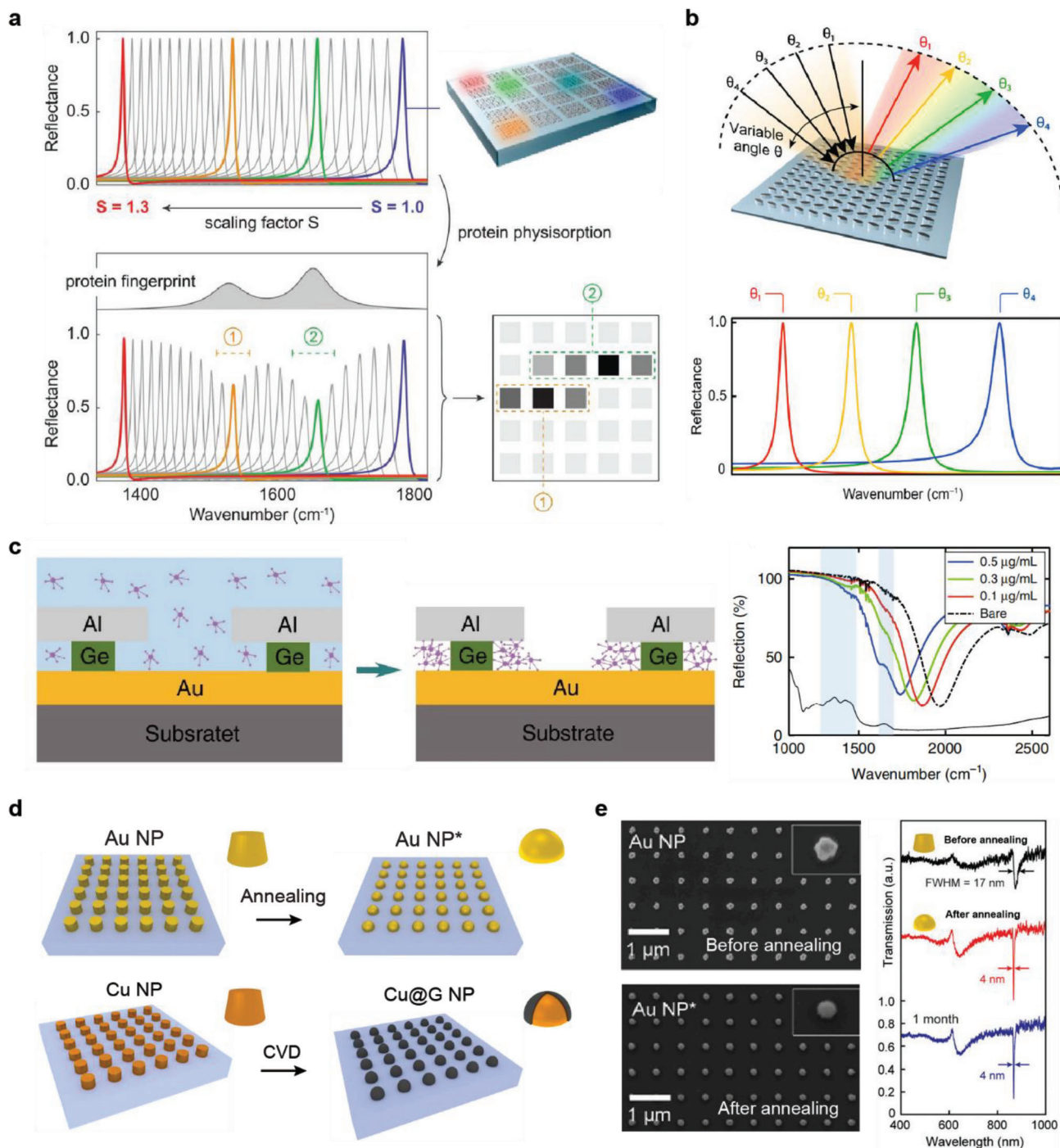


Figure 16. Device and fabrication innovation for sensing. a) Molecular fingerprint detection with pixelated hydrogenated amorphous Si metasurfaces; numerically simulated metapixel reflectance spectra (up) and example of molecular absorption fingerprint reproduced from reflectance amplitudes (bottom). b) A Ge-based high-Q all-dielectric metasurface that can support ultrasharp resonances at different incidence angles in broad spectral coverage. c) Schematic of Al trench array (top) and reflection spectra at different glucose solution concentrations (bottom). d) Schematic of annealing and chemical vapor deposition processes for high quality NP lattices. e) SEM images and transmission spectra of Au NP arrays before and after annealing. (a) Reproduced with permission.^[292] Copyright 2018, AAAS. (b) Reproduced with permission.^[297] Copyright 2019, AAAS. (c) Reproduced with permission.^[293] Copyright 2021, Springer Nature. (d) and (e) Reproduced with permission.^[125,298] Copyright 2020, NAS and Copyright 2022, NAS.

Table 2. Table of the acronyms in the text.

Terms	Acronyms
Surface plasmon polariton	SPP
Localized surface plasmon resonance	LSPR
Refractive index	RI
Nanoparticle	NP
Surface lattice resonance	SLR
Figure of merit	FOM
Full width at half maximum	FWHM
Metal-organic frameworks	MOFs
Limit of detection	LOD
Soft interface lithography	SIL
Polydimethylsiloxane	PDMS
Bovine serum albumin	BSA
Attenuated total reflection	ATR
Metal-dielectric-metal	MIM
Wood's anomaly	WA
Fabry–Pérot	FP
Surface-enhanced Raman scattering	SERS
Surface-enhanced infrared absorption	SEIRA
Cucurbit[7]	CB[7]
Part per billion	Ppb
Part per million	Ppm
Analytical enhancement factor	AEF
Volatile organic compounds	VOC
Single-stranded DNA	SsDNA
Polytetrafluoroethylene	PTFE
Poly methylmethacrylate	PMMA
Nanowire	NWR
Dinitrotoluene	DNT
Indium-tin-oxide	ITO
Aluminum-zinc-oxide	AZO
Gallium-zinc oxide	GZO
Mid-infrared	mid-IR
Transition metal dichalcogenides	TMDs
Quasi-bound states in the continuum	quasi-BICs
Circular dichroism	CD
Parity-time	PT

biosensing and gas sensing. In contrast to the conventional SPP and LSPR sensors, elaborately designed plasmonic architectures, such as diffractively coupled NP arrays, hybrid structures of plasmonic and functional materials, and nanostructures coupled with gain medium provide possibilities for ultrasensitive sensing down to the single molecule level. We also anticipate that advances in multiple aspects including detection mechanisms, innovations of devices, and advanced fabrication processes will further promote the development of highly sensitive sensing. Future research on plasmonic nanostructures will significantly improve sensitivity and selectivity towards high-performance sensing, and these prospects may also open opportunities for plasmonic nanostructures beyond sensing to novel nanophotonic de-

vices, bioimaging, quantum communication, and energy conversion (Table 2).

Acknowledgements

This work was supported by the National Natural Science Foundation of China (No. 62275257). The authors gratefully thank the startup funding from the Shanghai Institute of Microsystem and Information Technology, Chinese Academy of Sciences. The authors also thank Chang Guo for the help with the schemes, and Rui Ma and Rongrong Guo for proofreading.

Conflict of Interest

The authors declare no conflict of interest.

Keywords

hybrid metasurfaces, nanolasing, plasmonic lattices, selectivity, sensitivity

Received: February 16, 2023

Revised: April 7, 2023

Published online: May 19, 2023

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